

Infralimbic parvalbumin neural activity facilitates cued threat avoidance

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Abstract

The infralimbic cortex (IL) is essential for flexible behavioral responses to threatening environmental events. Reactive behaviors such as freezing or flight are adaptive in some contexts, but in others a strategic avoidance behavior may be more advantageous. IL has been implicated in avoidance, but the contribution of distinct IL neural subtypes with differing molecular identities and wiring patterns is poorly understood. Here, we study IL parvalbumin (PV) interneurons in mice as they engage in active avoidance behavior, a behavior in which mice must suppress freezing in order to move to safety. We find that activity in inhibitory PV neurons increases during movement to avoid the shock in this behavioral paradigm, and that PV activity during movement emerges after mice have experienced a single shock, prior to learning avoidance. PV neural activity does not change during movement toward cued rewards or during general locomotion in the open field, behavioral paradigms where freezing does not need to be suppressed to enable movement. Optogenetic suppression of PV neurons increases the duration of freezing and delays the onset of avoidance behavior, but does not affect movement toward rewards or general locomotion. These data provide evidence that IL PV neurons support strategic avoidance behavior by suppressing freezing.

eLife assessment

This **important** study extends our understanding of how the medial prefrontal cortex regulates flexible action during adversity. The data provide **compelling** evidence of a role for prefrontal PV neuron activity in active avoidance. This builds on the general idea that these neurons play a role in flexible behavior and demonstrates this in the context of freezing/avoidance conflict. The overall findings contribute to our understanding of mechanisms that support aversively motivated instrumental learning and may provide insight into both stress vulnerability and resilience processes. This work will be of interest to those interested in learning, aversive motivation, interneuron and/or prefrontal cortex function, or conditions relates to these processes and mechanisms.

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Introduction

The prefrontal cortex is essential for flexible behavior (Duncan, 1986; Miller and Cohen, 2001). Hallmarks of prefrontal damage in humans and animals include stimulus-bound and context-inappropriate behaviors, excessive reactivity, and impulsivity (Harlow, 1968; Bianchi and Macdonald, 1922; Lhermitte, 1983). The ventromedial part of the prefrontal cortex, the infralimbic cortex (IL), is important for supporting strategic behavior in the face of environmental threat (Murphy et al., 2005; Hardung et al., 2017). IL plays a critical role in fear extinction (Milad and Quirk, 2002; Do-Monte et al., 2015), discrimination between safety and fear (Sangha et al., 2014, 2020), and active avoidance (Moscarello and LeDoux, 2013; Halladay and Blair, 2017).

During active avoidance, mice first freeze in response to shock-predicting tones, as they do in fear conditioning, but gradually learn that they can avoid the shock by crossing the chamber when a tone plays. This behavior requires both the suppression of cued freezing and movement toward a safe zone (Mowrer and Lamoreaux, 1946; Koolhaas et al., 1999; Moscarello and LeDoux, 2013, 2014; Kryptos et al., 2015; LeDoux et al., 2017). Active avoidance is similar to fear extinction in that cue-elicited freezing behavior mediated by the amygdala is suppressed during both behaviors (Phillips and LeDoux, 1992; Kim et al., 1993; Herry et al., 2010; Milad and Quirk, 2012; Moscarello and LeDoux, 2013; LeDoux et al., 2017). IL neural activity is higher in rats that successfully extinguish freezing to conditioned stimuli, and IL projects directly and indirectly to the central amygdala and is thought to suppress central amygdala outputs that mediate freezing (Milad and Quirk, 2002; Vertes, 2004; LeDoux et al., 2017).

IL parvalbumin (PV) neurons synapse onto and inhibit local pyramidal neurons. Although we might expect that activation of IL PV neurons would inhibit avoidance behavior by inhibiting IL long-range projection neurons and disinhibiting central amygdala outputs that facilitate freezing, the relationship between PV and pyramidal neuron firing in cortex is complex. Cortical PV neurons have been reported to activate simultaneously with local pyramidal neurons (Merchant et al., 2008; Okun and Lampl, 2008; Isomura et al., 2009; Pinto and Dan, 2015; Estebanez et al., 2017; Nashef et al., 2022; Giordano et al., 2023), and it has been suggested that PV neurons may help to shape, rather than gate, the firing of local pyramidal neurons (Isomura et al., 2009; Merchant et al., 2012).

The question thus arises whether the activation of IL PV neurons suppresses or facilitates active avoidance behavior. Using fiber photometry, we show that IL PV neuron activity increases specifically when mice suppress cue-elicited freezing and move to avoid a future shock, but does not increase when animals move to obtain a cued reward or move in a neutral context. Further, we show that movement-related IL PV neural activity precedes avoidance learning, and emerges after mice have experienced a single shock in an environment, a finding that links PV neural activity specifically with the suppression of freezing to enable movement. Finally, we show that optogenetic suppression of IL PV neural activity prolongs freezing and delays avoidance but does not affect movement toward cued rewards or general locomotion. These results reveal that IL PV neurons play an essential and counterintuitive role in supporting flexible behavior in the face of threat, and suggest a role for PV neurons in shaping IL function that goes beyond the suppression of local neural activity.

Results

IL PV neurons signal active avoidance

We first asked how IL PV neurons respond during active avoidance behavior. To target this population for fiber photometry, we selectively expressed a genetically encoded calcium indicator in IL PV neurons by injecting AAV-CAG-Flex-GCaMP6f into IL in PV-Cre mice (Hippenmeyer et al., 2005 [↗](#); Chen et al., 2013 [↗](#)). In control mice we expressed GFP in IL PV neurons by injecting AAV-CAG-Flex-GFP. We implanted an optical fiber over IL to monitor calcium-dependent fluorescence, and recorded IL PV population activity via fiber photometry (**Figure 1A** [↗](#) and **B** [↗](#)) (Cui et al., 2013 [↗](#); Gunaydin et al., 2014 [↗](#)).

We used a two-way signaled active avoidance paradigm (**Figure 1C** [↗](#)). When an auditory cue (constant tone at 12 kHz or 8 kHz) was played, mice were required to cross the midline of the behavioral testing chamber within 5 seconds of tone onset to avoid an impending foot shock. If mice crossed the chamber during this 5-second period, no foot shock was delivered and the trial was scored as a successful avoidance. If not, a 2 second foot shock was delivered. Mice could terminate the foot shock early by crossing the chamber, which was scored as an escape. The tone terminated at either successful avoidance or shock offset. Prior to avoidance training, mice received two tone-shock pairings to learn the association between the auditory cue and the foot shock.

As IL inactivation impairs avoidance (Moscarello and LeDoux, 2013 [↗](#); Halladay and Blair, 2017 [↗](#)), we predicted that the activity of inhibitory IL PV neurons would be low during successful avoidance trials. Contrary to expectations, we found that IL PV neural activity rose upon the initiation of avoidance movements and peaked at chamber crossing (**Figure 1D-G**, **G** [↗](#): N = 8 mice, p=0.0077, paired t-test; **Figure 1—Figure supplement 1A-F**, **B** [↗](#): N = 8 mice, p = 0.0013, paired t-test, **F**: N = 8 mice, p = 0.4682, paired t-test). The trial-by-trial variability in the amplitude of IL PV neural activity was not correlated with variability in the speed or latency of the avoidance movement (**Figure 1—Figure supplement 1G-L**, **H** [↗](#): N = 8 mice, p = 0.3541, one-sample t-test, **K**: N = 8 mice, p = 0.4132, one-sample t-test). IL PV neural activity during the avoidance movement was not attenuated by learning or repeated reinforcement (**Figure 1—Figure supplement 1M** [↗](#) and **N** [↗](#), N = 8 mice, p = 0.8886, 1-way ANOVA).

IL PV neural activity reflects the avoidance movement, not the predictive tone

During successful avoidance trials, two events happen simultaneously: the mouse crosses the chamber, and a shock-predicting tone is terminated. To determine whether PV neural activity better reflects the avoidance movement or the termination of the predictive auditory sensory cue, we designed a version of the avoidance task with additional trial types to uncouple these events. Regular trials (80%) were interleaved with trials with shortened (10%) or lengthened (10%) tones (**Figure 2A** [↗](#)). In short-tone trials the tone lasted for only 1.5 seconds, and in long-tone trials the tone was not terminated until 1.5 seconds after successful avoidance (**Figure 2A** [↗](#)). IL PV neural activity at chamber crossing did not differ between regular and long-tone trials (**Figure 2B** and **D** [↗](#), N = 7 mice, p = 0.9546, paired t-test). Usually, the chamber was not crossed in short-tone trials, so short-tone trials were not included in this analysis. PV neural activity at tone offset was significantly different among short-tone, regular, and long-tone trials (**Figure 2C** and **E** [↗](#), N = 7 mice, p = 0.005, one-way repeated ANOVA. Multiple comparison, R and S, p=0.0045, R and L, p=0.0398, S and L, p=0.4620.). These results indicate that PV neural activity primarily reflects the avoidance movement and not tone offset.

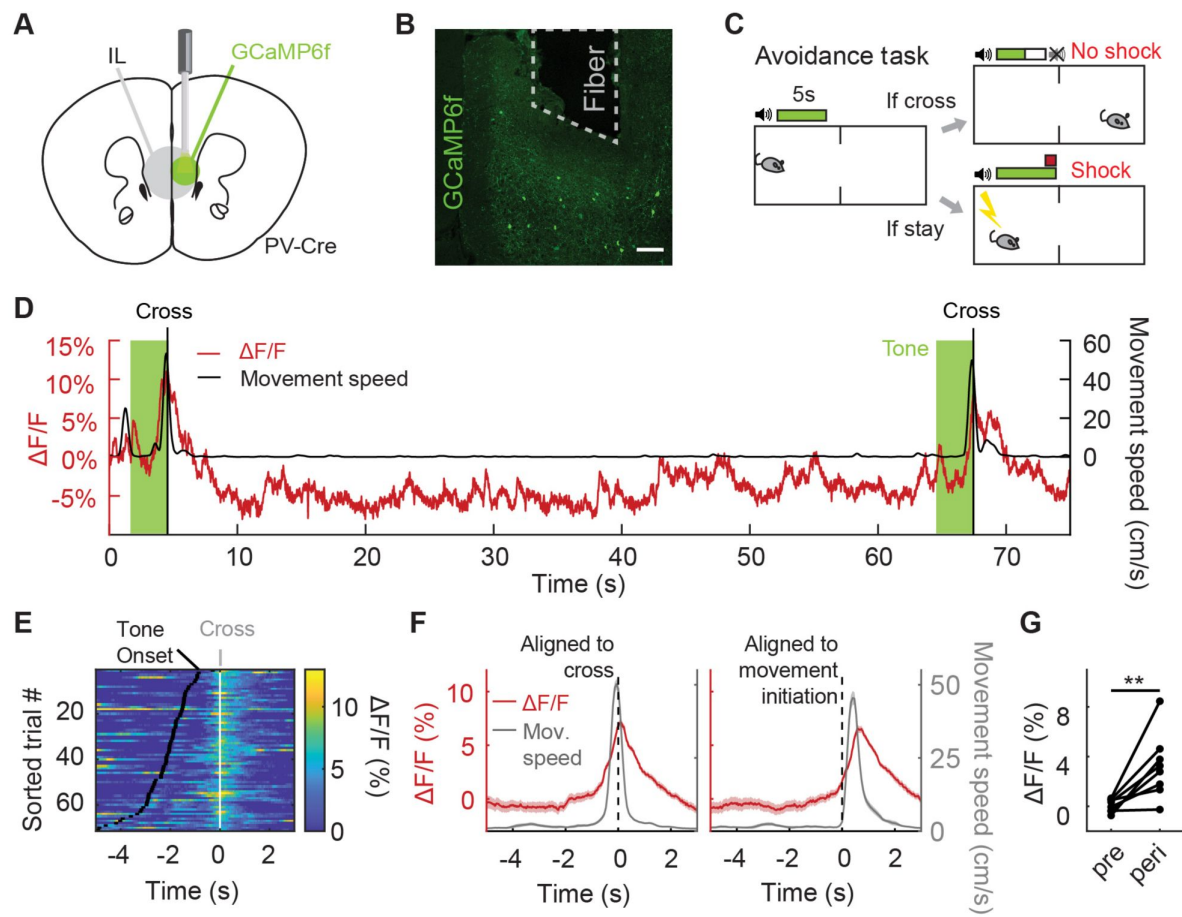


Figure 1.

IL PV neurons signal active avoidance

(A) Fiber photometry schematic. (B) GCaMP6f expression in IL PV neurons. Scale bar, 100 μ m. (C) Active avoidance task schematic. The green box indicates presence of the tone, which lasts till animal crosses. (D) Example IL PV $\Delta F/F$ (red) and speed (black) during two successful avoidance trials. Vertical line indicates chamber crossing. Tone, light green. (E) IL PV $\Delta F/F$ during successful avoidance trials, data aligned to chamber crossing. White ticks: chamber crossing. Black ticks: tone onset. Same example mouse as D. (F) Example average IL PV $\Delta F/F$ (red) and speed (grey), aligned to chamber crossing (left) and movement initiation (right). Same example mouse as D. (G) Average IL PV $\Delta F/F$ before chamber crossing (pre, 4 to 2 seconds before cross) and during chamber crossing (peri, 1 second before to 1 second after crossing). ** $p < 0.01$, paired t-test. Shaded regions indicate SEM.

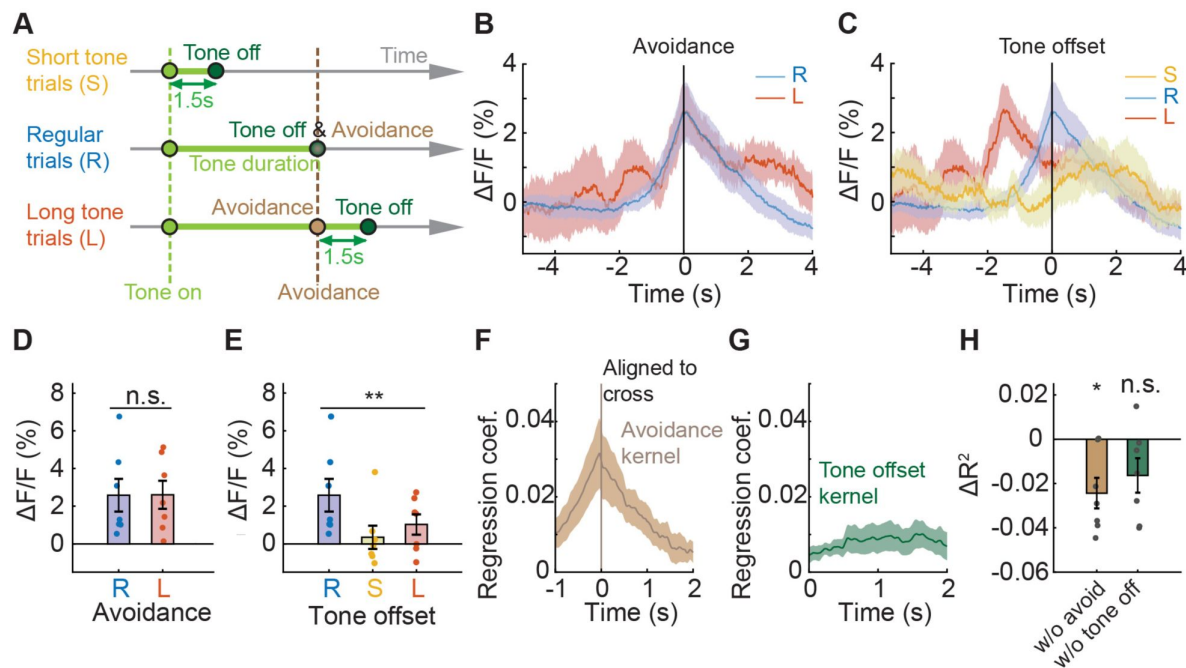


Figure 2.

IL PV neural activity reflects the avoidance movement, not the predictive tone

(A) Schematic of a modified version of avoidance task including 10% short tone trials (S), where the tone is always 1.5 seconds; 10% long tone trials (L), where the tone lasts until 1.5 seconds after successful avoidance; and 80% regular trials (R), where the tone terminates upon successful avoidance. (B-C) Average IL PV $\Delta F/F$ aligned to (B) chamber crossing and (C) tone offset. (D) Comparison between average IL PV calcium activity 0 to 0.1 seconds after avoidance chamber crossings during regular trials and long tone trials (ns = non-significant, paired t-test). (E) Comparison between average IL PV calcium activity 0 to 0.1 seconds after tone offsets during regular trials, short tone trials, and long tone trials (**p < 0.01, one-way repeated ANOVA). (F-G) Average avoidance kernel (F) and average tone offset kernel (G) across all animals calculated by a linear model. (H) Loss of predictive power (ΔR^2) in a reduced model with shuffled avoidance or tone offset time points (ns = non-significant, *p < 0.05, one-sample t test). Error bars and shaded regions indicate SEM.

We extracted characteristic neural responses to each event without interference from other events close in time. We constructed a linear regression model to extract the isolated neural responses to chamber crossing and tone termination (Parker et al., 2016 [DOI](#); Musall et al., 2019 [DOI](#)), and all major events were included in the model. We assumed (1) that neural responses would be similar for the same events and dissimilar for different events, and (2) neural responses to events can be summed up linearly to form the recorded signals. With these assumptions and through linear regression, isolated neural responses to each event were extracted. The model-extracted isolated neural response to tone offset was flat, while the isolated avoidance signal peaked at chamber crossing (Figure 2F and G [DOI](#)). To further demonstrate that IL PV activity can be better accounted for by the action to avoid the shock than tone termination, we compared the explanatory power (R^2 , coefficient of determination) of the reduced model with shuffled time points to the full model. Shuffling the avoidance time points significantly reduced the explanatory power of the model (ΔR^2 different from zero, $N = 7$ mice, $p = 0.0125$, one-sample t-test), while shuffling the tone offset time points had no effect (Figure 2H [DOI](#), $N = 7$ mice, $p = 0.0802$, one-sample t-test). Thus, IL PV neural activity at chamber crossing reflects the avoidance behavior and not the cessation of the predictive auditory cue.

IL PV neural activity does not reflect movement to obtain rewards

IL PV neural activity rises during movement to avoid a predicted future shock (Figure 1E and F [DOI](#)), but it is unclear whether this neural activity specifically reflects avoidance, or if IL PV neural activity would be elevated during any movement. If IL PV neural activity reflects locomotor activity, we would expect to see elevated neural activity during movement to obtain rewards. To test this idea, we recorded IL PV neuronal activity during a reward approach task, which mirrored the avoidance design in temporal structure, but shock omission on successful chamber crossing was replaced with reward delivery (Figure 3A [DOI](#)). In this task, chamber-crossing movements to obtain a water reward were not associated with elevated IL PV neural activity (Figure 3B-E; E [DOI](#), $N = 7$ mice, $p = 0.3003$, paired t-test; Figure 3—Figure supplement 1A-C, C [DOI](#); $N = 6$ mice, $p = 0.0382$, paired t-test; see Figure 1C-G [DOI](#) for comparison).

We observed an elevation of IL PV neuronal activity after chamber crossing when the animals started to consume the water reward (Figure 3C [DOI](#), magenta dots, and Figure 3D [DOI](#)). To test whether this transient increase in IL PV neuron activity after chamber crossing reflects water consumption or the preceding approach behavior, we introduced randomized water-omission trials in 10% of the trials (Figure 3—Figure supplement 1D-E [DOI](#)). We found that the PV neuron signal rose both when water was received and when the water reward was omitted, and increases in IL PV activity preceded the first lick to consume water reward delivered after a successful crossing (Figure 3—Figure supplement 1E [DOI](#)). We speculate that, rather than a reward signal related to water consumption, this signal may reflect the suppression of the habitual chamber-crossing movement in the approach task when the mouse nears the lick port.

To further investigate whether elevated IL PV neuronal activity reflected avoidance or locomotion, we recorded IL PV activity in an open field test (OFT) where animals were allowed to run freely without any defined task structure (Figure 3—Figure supplement 2A [DOI](#) and B [DOI](#)). When we aligned IL PV activity to movement peaks, no elevated activity was observed (Figure 3—Figure supplement 2C [DOI](#)). We varied the overhead lighting and found that IL PV neurons showed no elevated activity during movement under either a bright ceiling light, which is a more aversive setting to the mice, or a dim ceiling light, which is more comforting to the mice (Figure 3—Figure supplement 2D [DOI](#) and E [DOI](#), $N = 7$ mice, bright light: $p = 0.8705$, dim light: $p = 0.1475$, paired t-test; Figure 3—Figure supplement 2F [DOI](#)). It should be noted that the OFT is a relatively neutral context, regardless of lighting conditions, when compared to the appetitive reward approach or aversive active avoidance.

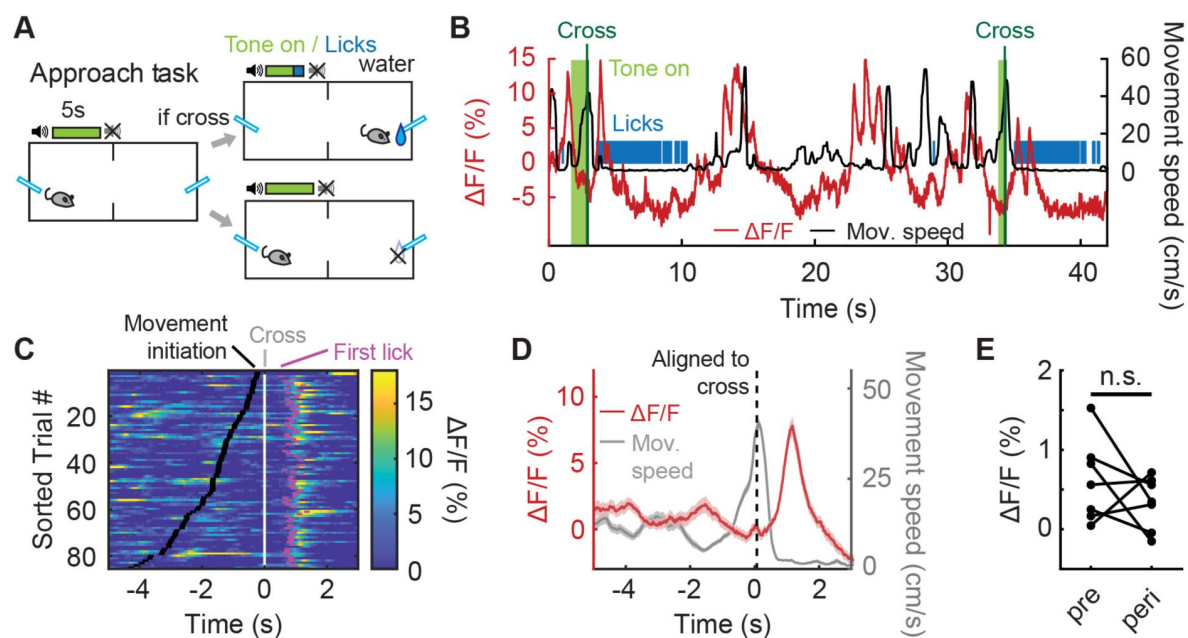


Figure 3.

IL PV neural activity does not reflect movement to obtain rewards.

(A) Reward approach task schematic. (B) Example IL PV $\Delta F/F$ (red) and speed (black) during two successful approach trials. Vertical line indicates chamber crossing. Tone, light green. (C) IL PV $\Delta F/F$ during successful approach trials, data aligned to chamber crossing. Black ticks: tone onset; white ticks: chamber crossing; magenta ticks: first licks. Same example mouse as B. (D) Example average IL PV $\Delta F/F$ (red) and speed (grey), aligned to chamber crossing (left). Same example mouse as B. (E) Average IL PV $\Delta F/F$ before chamber crossing (pre, 4 to 2 seconds before cross) and during chamber crossing (peri, 1 second before to 1 second after crossing). ns = non-significant, paired t-test. Shaded regions indicate SEM.

IL PV neural activity becomes positively correlated to movement after shock

In environments with different emotional valence animals engage in different suites of behaviors. For example, animals in threatening environments spend more time freezing and less time exploring than animals in rewarding environments. We hypothesized that IL neural activity may become positively correlated with movement when animals learn that their environment is threatening, and must suppress behaviors such as freezing in order to engage in behaviors such as exploration.

To test this hypothesis, we analyzed the recordings made on the first day of active avoidance training to determine how movement-related IL PV neural activity evolves as animals learn that the environment is threatening. On the first session, we habituated animals to the chamber for 5 minutes before the task started. During this habituation period, IL PV activity was not positively correlated with movement ([Figure 4A](#); [Figure 4—Figure supplement 1A](#)), similar to our observations in appetitive and neutral contexts ([Figure 3](#); [Figure 3—Figure supplement 1E-I](#)). During avoidance behavior, we found that IL PV activity peaked during movements in the inter-trial interval ([Figure 4B](#); [Figure 4—Figure supplement 1A](#)). Considering that animals minimize their movements in threatening environments, movement outside of the tone-evoked trial could require suppression of this natural inclination.

The correlation between movement and IL PV neural activity during the inter-trial interval did not emerge until after animals experienced the first foot shock ([Figure 4C](#); [Figure 4—Figure supplement 1A](#)). The correlation between movement and IL PV neuronal activity emerged immediately after receiving the first shock and did not increase during the learning process, suggesting that avoidance learning is not necessary for the correlation between movement and IL PV neural activity to emerge. The lack of positive correlation between activity and movement in the habituation phase ([Figure 4A-C](#); [Figure 4—Figure supplement 1A](#)) of the avoidance task is similar to what was observed throughout the approach task ([Figure 4D](#); [Figure 4—Figure supplement 1B](#)). These data show that the positive correlation between movement and PV neuron activity emerges immediately after animals learn that their current environment is threatening, when movement requires a suppression of freezing.

Inhibiting IL PV neural activity delays avoidance

To investigate the causal role of IL PV neural activity in active avoidance, we bilaterally inhibited IL PV neurons by expressing Cre-dependent halorhodopsin (AAV5-EF1 α -DIO-eNpHR; control animals: AAV5-EF1 α -DIO-eYFP) in PV-Cre mice ([Figure 5A](#) and [5B](#)). We used the same avoidance and approach tasks described above ([Figure 1A](#); [Figure 3A](#)), and introduced interleaved simulation blocks where IL PV neurons were optogenetically inhibited during the interval from 0.5 to 2.5 seconds after tone onset ([Figure 5C](#)). Inhibiting IL PV neuronal activity delayed the avoidance movement ([Figure 5D-G](#); [Figure 5—Figure supplement 1A](#) and [B](#); [Video 1](#)) but did not delay the movement for water reward in the approach task ([Figure 5H-K](#); [Figure 5—Figure supplement 1C](#) and [D](#)) or the speed of voluntary locomotion in the OFT ([Figure 5L-O](#), dim OFT: interaction between opsins and stimulation: $N = 8$ NpHR mice, $N = 5$ eYFP mice, $p = 0.7636$, 2-way ANOVA; bright OFT: interaction between opsins and stimulation: $N = 8$ NpHR mice, $N = 5$ eYFP mice, $p = 0.7469$, 2-way ANOVA).

Both the speed of the avoidance movement ([Figure 5D](#) and [5E](#), significant interaction between opsin and stimulation, $N = 8$ NpHR mice, $N = 5$ eYFP mice, $p = 0.0281$, 2-way ANOVA) and the probability of successful avoidance ([Figure 5F](#) and [5G](#), $N = 8$ NpHR mice, $N = 5$ eYFP mice, $p = 0.0058$, unpaired t-test) were reduced by suppression of IL PV neurons. The speed of movement and the probability of successful reward delivery were not affected in the reward approach task (speed: [Figure 5H](#) and [5I](#), interaction between opsin and stimulation, $N = 8$ NpHR mice, $N = 5$

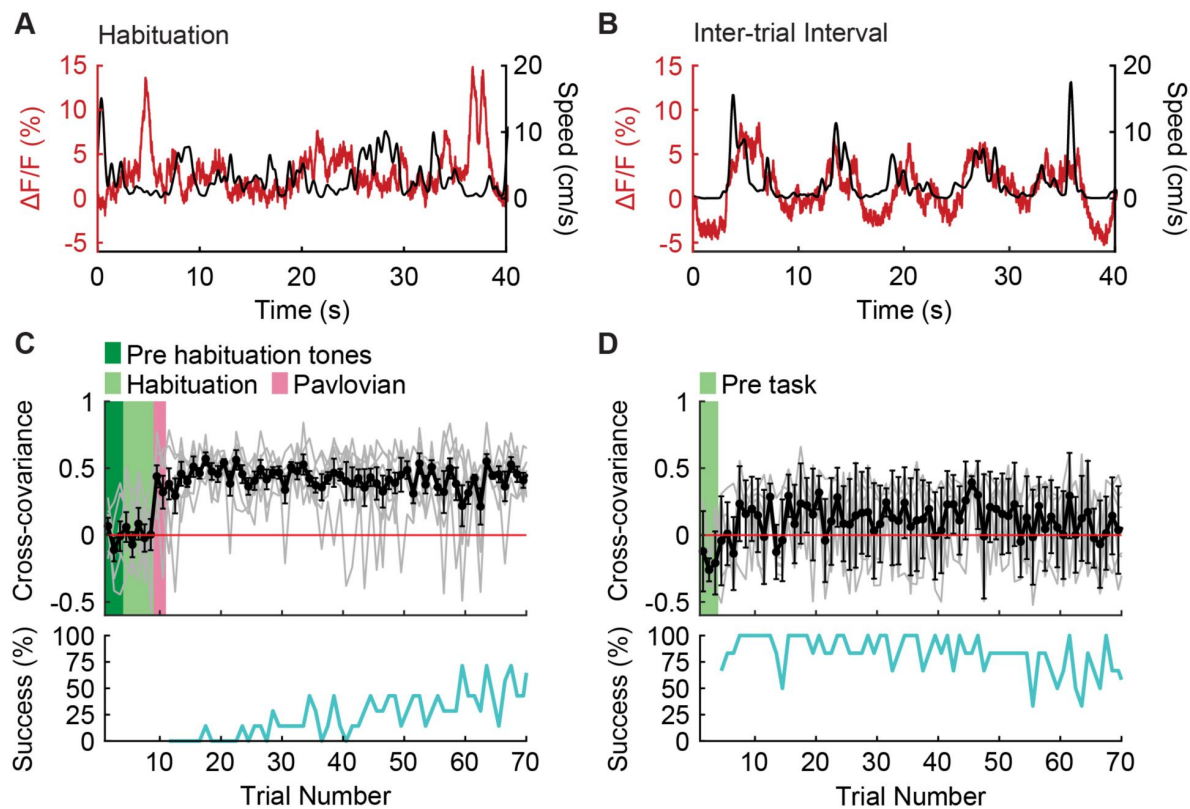


Figure 4.

IL PV neural activity becomes correlated to movement after shock.

(A-B) Example IL PV $\Delta F/F$ (red) and speed (black) during (A) habituation prior to the first-ever shock exposure, and (B) the intertrial interval after avoidance training. (C-D) Evolution of cross-covariance between IL PV activity and speed over trials (upper panel, black) and corresponding average avoidance success rate across animals (bottom panel, cyan) (C) on the first day of avoidance training (N = 7 mice) and (D) during approach in well-trained mice (N = 6 mice, green shading marks the time in chamber before the task (pre-task)). Error bars indicate SEM.

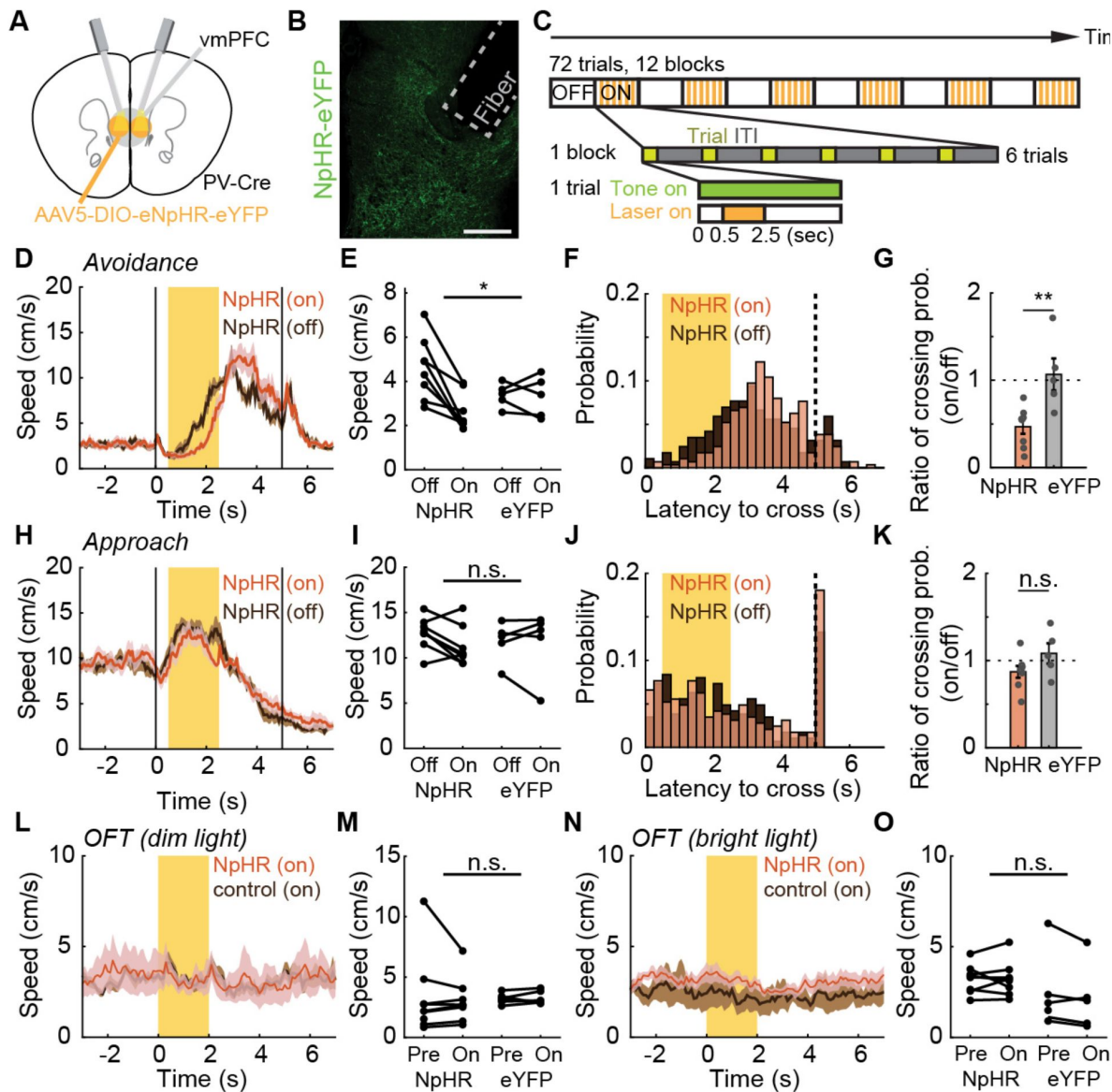


Figure 5.

Inhibiting IL PV neural activity delays avoidance.

(A) Optogenetic schematic. (B) NpHR-eYFP expression in IL PV neurons in a PV-Cre mouse. Scale bar: 200 μ m. (C) Optogenetic inactivation schematic. Interleaved stimulation blocks were introduced, with optogenetic inhibition of IL PV neurons 0.5 to 2.5 seconds after tone onset. (D) Speed of NpHR-expressing mice during avoidance with (red) and without (dark brown) illumination. (E) Speed of NpHR- and eYFP-expressing mice, averaged over the laser stimulation period, with and without illumination. (F) Distribution of crossing latencies during avoidance in NpHR-expressing animals with (red) and without illumination (brown). (G) Ratio of illuminated/non-illuminated chamber crossing probabilities. (H-K) Same as (D-G) for approach task. (L-M) Same as (D-E) but for OFT with dim light. (N-O) Same as (L-M) but for OFT with bright light. ns = nonsignificant, * $p < 0.05$, ** $p < 0.01$; for (E), (I), (M), and (O), 2-way ANOVA interaction term; for (G) and (K), unpaired t-test. Shaded regions indicate SEM.

eYFP mice, $p = 0.479$, 2-way ANOVA; probability of approach: **Figure 5J** and **5K**, $N = 8$ NpHR mice, $N = 5$ eYFP mice, $p = 0.1204$, unpaired t-test). This suggests that IL PV neuronal activity is not just correlated with avoidance behavior but plays a causal role. By further analyzing the video data, we found that freezing during tone presentation increased (**Figure 5—Figure supplement 1E–G**, **G**: $N = 8$ NpHR mice, $N = 5$ eYFP mice, $p = 0.0175$, unpaired t-test; **Video 1**). In addition, the ratio of freezing duration to avoidance latency increased during IL PV inhibition (**Figure 5—Figure supplement 1H**, $N = 8$ NpHR mice, $N = 5$ eYFP mice, $p = 0.0302$, unpaired t-test; **Video 1**), suggesting that increased avoidance latency was primarily due to an increase in freezing. This finding supports the idea that IL PV neural activity promotes avoidance by suppressing freezing (Murphy et al., 2005; Hardung et al., 2017).

We then asked whether inhibiting IL PV neuronal activity had an immediate effect on avoidance during the current trial or if instead inhibition influenced learning. To address this question, we investigated whether IL PV inhibition affected avoidance in trials following inhibition, using the task described in **Figure 5C**. Contrary to the learning hypothesis, we found that suppressing IL PV neurons delayed avoidance even on the first trial of a stimulation block. The latency of the first trial of inhibition blocks was significantly longer than the latency of pre-stimulation trials (**Figure 5—Figure supplement 1I** and **J**, $N = 8$ NpHR mice, $N = 5$ eYFP mice, $p = 0.0022$, paired t test), but was not significantly different from the latency of subsequent inhibition trials (**Figure 5—Figure supplement 1I** and **J**, $N = 8$ NpHR mice, $N = 5$ eYFP mice, $p = 0.0943$, paired t test). The latency of the first non-inhibited trial following an inhibition block is significantly different from the latency of the preceding inhibition trials (**Figure 5—Figure supplement 1K** and **L**, $N = 8$ NpHR mice, $N = 5$ eYFP mice, $p = 0.0343$, paired t test) but is not significantly different from the latency of subsequent non-inhibited trials (**Figure 5—Figure supplement 1K** and **L**, $N = 8$ NpHR mice, $N = 5$ eYFP mice, $p = 0.8147$, paired t-test). These results suggest that IL PV neuronal activity plays a causal role in modulating ongoing behavior.

To further tease apart the role of IL PV neural activity in learning, we inhibited IL PV neurons immediately after successful chamber crossing in the avoidance task, rather than during the tone (**Figure 5—Figure supplement 1M**). With this experimental design, speed was not affected by IL PV inhibition (**Figure 5—Figure supplement 1N–P**, **P**: $N = 8$ NpHR mice, $N = 5$ eYFP mice, $p = 0.8846$, 2-way ANOVA interaction term). The latency of the first trials in inhibition blocks showed no significant difference from the latency of the second trials in inhibition blocks (**Figure 5—Figure supplement 1Q**, $N = 8$ NpHR mice, $N = 5$ eYFP mice, $p = 0.6944$, paired t-test), and the latency of the first trials after the end of the inhibition block also showed no difference to the latency of the second trials after the inhibition block (**Figure 5—Figure supplement 1Q**, $N = 8$ NpHR mice, $N = 5$ eYFP mice, $p = 0.6745$, paired t-test). Thus, post-avoidance inhibition of IL PV neural activity did not affect avoidance latency in subsequent trials.

Discussion

Here, we show that IL PV neural activity rises during movement to avoid a cued future shock, but not during movement to obtain reward or movement in the open field. Further, this rise in activity during movement emerges immediately after the first shock and does not require avoidance learning. We also show that inhibiting IL PV neurons prolongs freezing and delays avoidance, but does not affect movement to obtain reward or movement in the open field. These results demonstrate that IL PV neurons play a key role in suppressing freezing in order to permit movement in threatening environments, an essential component of the avoidance response.

The rise in IL PV neural activity during movement does not require avoidance learning – IL PV neurons begin to respond during movement immediately after the animal has received a single shock in an environment, but learning to cross the chamber to avoid the signaled shock takes tens of trials. Why is there a discordance between the emergence of the IL PV signal during movement

and avoidance learning? The components underlying active avoidance have been debated over the years, but are thought to involve at least two essential behaviors – suppressing freezing, and moving to safety (LeDoux et al., 2017 [↗](#)). Freezing is the default response of mice upon hearing a shock-predicting tone, and can be learned in a single trial (Ledoux, 1996 [↗](#); Fanselow, 2010 [↗](#); Zambetti et al., 2022 [↗](#)). When a predator is in the distance, freezing can increase the chance of survival by reducing the chances of detection. However, a strategic avoidance behavior may prevent a future encounter with the predator altogether. The importance of IL PV neural activity in defensive behavior may be to suppress reactive defensive behaviors such as freezing in order to permit a flexible goal-directed response to threat.

The freezing suppression and avoidance movement components of the avoidance response are dissociable, both because freezing precedes avoidance learning, and because animals intermittently move prior to avoidance learning. Our finding that the rise in PV activity during movement emerges immediately after receiving a single shock, tens of trials before animals have learned the avoidance behavior, suggests that the IL PV signal is associated with the suppression of freezing. Further, IL PV neurons do not respond during movement toward cued rewards because in reward-based tasks there is no freezing response in conflict with reward approach behavior.

We think the IL PV signal is unlikely to be a safety signal (Sangha et al., 2020 [↗](#)). First, the PV signal rises during movement not only in the avoidance context, but during any movement in a “threatening” context (i.e. a context where the animal has been shocked). For example, PV neural activity rises during movement during the intertrial interval in the avoidance task. Further, the emergence of the PV signal during movement happens quickly – after the first shock – and significantly before the animal has learned to move to the safe zone. This suggests a close association with enabling movement in a threatening environment, when animals must suppress a freezing response in order to move. Additionally, the rise in PV activity was specifically associated with movement and not with tone offset, the indicator of safety in this task. Finally, if IL PV neural activity reflects safety signals one would expect the response to be enhanced by learning, but the amplitude of the IL PV response was unaffected by learning after the first shock.

Finding that inhibitory IL PV neural activity suppresses freezing was counterintuitive, given the importance of IL for fear extinction and avoidance learning. We had predicted that IL PV neurons would be suppressed during avoidance, because IL is active when animals have successfully extinguished freezing in response to shock-predicting cues (Milad and Quirk, 2002 [↗](#)), and muscimol inhibition of IL impairs avoidance learning (Moscarello and LeDoux, 2013 [↗](#)). However, recent studies have suggested that the role of cortical PV neurons goes beyond suppressing overall neural activity in a region, and likely plays a more delicate role in tuning local computations.

For example, PV neurons aid in improving visual discrimination through sharpening response selectivity in visual cortex (Lee et al., 2012 [↗](#)). In prefrontal cortex, PV neurons are critical for task performance, particularly during performance of tasks that require flexible behavior such as rule shift learning (Cho et al., 2020 [↗](#)) and reward extinction (Sparta et al., 2014 [↗](#)). Further, PV neurons play an essential role in the generation of cortical gamma rhythms, which contribute to synchronization of selective populations of pyramidal neurons (Sohal et al., 2009 [↗](#); Cardin et al., 2009 [↗](#)). Courtin et al (2014) [↗](#) showed that brief suppression of dorsomedial prefrontal (dmPFC) PV neural activity enhanced fear expression, one of the main functions of the dmPFC, by synchronizing the spiking activity of dmPFC pyramidal neurons (Courtin et al., 2014 [↗](#)). This result is potentially relevant to our findings, but likely involves different circuit mechanisms because of the difference in timescale, targeted area, and downstream projection targets (Vertes, 2004 [↗](#)). These and other studies support the idea that PV neural activity supports the execution of a behavior by shaping rather than suppressing cortical activity, potentially by selecting among conflicting behaviors by the synchronization of different pyramidal populations (Warden et al., 2012 [↗](#); Lee et al., 2014 [↗](#)). The roles of other inhibitory neural subtypes (such as somatostatin

(SOM)-expressing and vasoactive intestinal peptide (VIP)-expressing IL GABA neurons) in avoidance behavior are currently unknown, but are likely important given the role of SOM neurons in gamma-band synchronization (Veit et al., 2017 [DOI](#)), and the role of VIP neurons in regulating PV and SOM neural activity (Cardin, 2018 [DOI](#)).

Our findings have revealed the role of IL PV neural activity in facilitating flexible avoidance behavior by suppressing the conflicting freezing behavior. Though IL PV neurons comprise only a relatively sparse cortical population, inhibiting these neurons has a clear and specific detrimental effect on the initiation of avoidance behavior, which is vital for leaving a dangerous situation. Our work also suggests that the functional role of PV neurons extends beyond the overall suppression of the function of a brain region, and provides a conceptual framework of potential utility for deepening our understanding of the functional roles of PV neurons in mediating conflict between behaviors through coordination of local circuits.

Materials and methods

Animals

All procedures conformed to guidelines established by the National Institutes of Health and have been approved by the Cornell University Institutional Animal Care and Use Committee. PV-Cre mouse line (B6.Cg-Pvalb^{tm1.1(cre)Aib}/J, RRID:IMSR_JAX:012358) acquired from The Jackson Laboratory (Bar Harbor, ME) was backcrossed to C57BL/6J mice (RRID:IMSR_JAX:000664). Postnatal six weeks to ten months old PV-Cre male mice were used for all photometry and optogenetics experiments. All mice were housed in a group of two to five under 12-hour reverse light-dark cycle (dark cycle: 9 a.m.-9 p.m.).

Viral Vectors

In the photometry experiment, we used AAV1-CAG.Flex.GCaMP6f (Titer: 1.33×10^{13} , Penn Vector Core, 100835-AAV1, Philadelphia, PA) for experimental animals and AAV9-CAG.Flex.GFP (Titer: 3.7×10^{12} , UNC Vector Core, Chapel Hill, NC) for control. All the viral vectors used in photometry experiment were diluted in 8-fold phosphate-buffered saline (PBS) before injection. In the optogenetics experiment, we used AAV5-EF1 α -DIO-eNPHR3.0 (Titer: 4×10^{12} , UNC Vector Core, Chapel Hill, NC) for experimental animals and AAV5-EF1 α -DIO-EYFP (Titer: 6.5×10^{12} , UNC Vector Core, Chapel Hill, NC) for control. All the viral vectors used in optogenetics experiment were used without dilution.

Surgical Procedures

Mice were put under deep anesthesia with isoflurane (5%). Fur above the skull was trimmed, and the mice were placed in a stereotaxic frame (Kopf Instrument, Tujunga, CA) with a heating pad to prevent hypothermia. Isoflurane level was kept between 0.8 to 2% throughout the surgery. Ophthalmic ointment was applied to protect the eyes. 100 μ L 1 mg/ml Baytril (enrofloxacin) was given subcutaneously, and 100 μ L 2.5 mg/ml bupivacaine was injected subdermally at the incision site. The scalp was disinfected with betadine and alcohol. The skull was exposed with a midline incision. A craniotomy was made above the medial prefrontal (mPFC) cortex.

For fiber photometry animals, virus (AAV1-CAG-Flex-GCaMP6f) was injected into mPFC unilaterally, with half of the animals into the right hemisphere and half of the animals into the left hemisphere (infralimbic cortex (IL) coordinates: -1.55 AP, ± 0.3 ML, -2.8 to -3.2 DV). A total of 800 nl diluted vector (1:8 dilution) was injected in each mouse. Virus injection was done with a 10 μ L Hamilton syringe (nanofil, WPI, Sarasota, FL) and a 33-gauge beveled needle, and a micro-syringe pump controller (Micro 4; WPI, Sarasota, FL) using slow injection speed (100 nl/min). The needle

was slowly withdrawn 15 minutes after injection. After injection, a 4 mm or 6 mm-long optic fiber (diameter: 400 μm , 0.48NA, Doric Lenses, Quebec, Canada) was implanted 0.5 to 1 mm above the injection site.

For optogenetics animals, virus (AAV5-EF1a-DIO-eNpHR3.0, Lot #: 4806G, titer: 4.00×10^{12} ; control: AAV5-EF1a-DIO-eYFP, Lot #: 4310J, titer: 6.5×10^{12} , UNC vector core, NC, USA) was injected into mPFC bilaterally (IL coordinates: -1.55 AP , $\pm 0.3 \text{ ML}$, -2.8 to -3.2 DV). 500 nl vector was injected into each site on each mouse. After injection, a 4 mm to 6 mm-long optic fiber (diameter: 200 μm , 0.22NA, Thorlabs, NJ, USA) was implanted 0.5 to 1 mm above the injection site in a 15-degree angle toward midline ($\text{AP} = 1.55$, $\text{ML} = \pm 1.05$ or 1.15 , $\text{DV} = -2.8$ or -2.99 , 15-degree angle).

After the fiber implant, a layer of metabond (Parkell, Inc., Edgewood, NY) and dental acrylic (Lang Dental Manufacturing, Wheeling, IL) was applied to hold the implant in place. Buprenorphine (0.05 mg/kg), carprofen (5 mg/kg), and lactated ringers (500 μL) were administered subcutaneously after surgery. Photometry recording was done no earlier than three weeks later to allow for virus expression.

Fiber Photometry

Fiber photometry was implemented with a fiber photometry console (Doric Lenses, Quebec, Canada). An FC-FC optic fiber patch cord (400 μm diameter, 0.48NA, Doric Lenses, Quebec, Canada) was connected to implanted fiber with a zirconia sleeve. 405 nm and 475 nm were measured for calcium-independent and calcium-dependent GCaMP signals and were measured with digital lock-in frequency to the input at 208Hz and 530Hz, respectively. Photometry signals were collected at 12 kHz and filtered with a 12 Hz low pass filter.

For photometry animals, four GCaMP6f animals were tested only on active avoidance. Seven GCaMP6f animals and two GFP control animals were tested in the following order: open field test, reward approach test, reward approach with omission (except two GFP and two GCaMP6f animals), active avoidance, active avoidance with shortening and extended tone, active avoidance extinction.

Open Field Test (Photometry)

A 46 W x 46 L x 30.5 H (cm) white rectangular box made with PVC was used for an open field test. The ambient light was set for each 2-minute block in the order of D(dim)-B(bright)-D-B-D-B-D-B-D-B.

Reward Approach (Photometry)

The reward approach task was performed in a 17.25" W x 6.75" D x 10" H metal rectangular shuttle box (MedAssociates, Fairfax, VT) divided into two equal compartments by Plexiglas semi-partitions, which allowed animals to move freely between compartments. A water sprout was located at the end of each compartment, and a syringe pump was connected to the sprout for water delivery. Licks were detected with a contact-based lickometer (MedAssociates, Fairfax, VT). Animals were water restricted prior to training. Body weight was checked daily and was maintained above 80% baseline. Animals were trained to first learn the association between tone and reward-licking at the waterspout in the opposite compartment, by playing a tone for an indefinite length of time (12 kHz or 8 kHz at 70-80 dB, counterbalancing between approach and avoidance) until successful reward collection. When an animal crossed the chamber in response to a tone, water was delivered in the goal compartment, and an indicator light above the targeted water sprout was terminated. After two to three days of training, the reward-indicating lights were removed. Tone duration was set to be turned off either at the chamber crossing or at maximum duration. Animals had to cross within the maximum duration of tone for successful water delivery. The maximum tone duration was shortened as animals progressed and were eventually

set to 5 seconds. The inter-trial interval was pseudo-randomized at an average of 40 seconds. Animals were allowed to perform 30-50 trials each day during training and 100 trials on the recording day. The training lasted two to three weeks until animals reached a 70% success rate with a 5-second window. Animals were recorded the day after criterion was reached for 100 trials.

Reward Approach with 10 % Omission (Photometry)

After training and recording in reward approach, animals were then switched to a 10% omission paradigm, where water was not delivered on 10% of the successful crossing trials. The omission trials were selected pseudo-randomly at a 10% chance.

Active avoidance (Photometry)

Active avoidance was performed in a 14" W x 7" D x 12" H metal rectangular shuttle box (Coulbourn Instruments, Holliston, MA) divided into two equal compartments by Plexiglas semi-partitions. Animals were habituated to the chamber for at least 30 minutes, the day before training. On the first day, animals received five five-second habituation tones and two Pavlovian conditionings (a seven-second tone and a two-second shock at the last two-seconds of tone) prior to avoidance trials. The same frequency and amplitude of tone were used for habituation tones, Pavlovian tones, and avoidance tones (12 kHz or 8 kHz at 70-80 dB, counterbalancing between approach and avoidance). After habituation and Pavlovian trials, the task was switched to avoidance trials where animals could prevent the shock by crossing the chamber within 5 seconds from the onset of the tone. Otherwise, an electrical foot shock (0.3 mA) would be delivered through the grid floor for a 2-second maximum before the animal crossed the chamber to escape the shock. The tone was terminated when animals crossed the chamber or after 7 seconds. The inter-trial interval was pseudo-randomized for an average of 40 seconds. Animals performed 100 avoidance trials per day for two to three days or until reaching 70% successful rate. Photometry data was recorded during training.

Active avoidance with 10% shortened and extended tones (Photometry)

After active avoidance recording, animals were then recorded in an alternative version of active avoidance with 10% shortened and extended tones. The only change in this alternative version was to include 10% trials with shortened tones which were turned off after 1.5 seconds regardless, and 10% trials with extended tones, which were extended for 1.5 seconds after successful avoidance. The shortened and extended trials were selected pseudo-randomly at 10% chance for each trial. No shock was delivered in any shortened-tone and extended-tone trials.

Optogenetics

In the behavioral experiment, two external FC-FC optic fiber patch cords (200 μ m diameter, 0.22 NA, Doric Lenses, Quebec, Canada) were connected to two implanted fibers, respectively, each with a zirconia sleeve. These patch cords were then connected to a 1x2 fiber-optic rotary joint (FRJ_1x2i_FC-2FC_0.22, Doric Lenses, Quebec, Canada) for unrestricted rotation and to prevent tangling. Another FC-FC optic fiber patch cord was used to connect the rotary joint to a 100 mW 594 nm diode-pumped solid-state laser (Cobolt MamboTM 100 594nm, HÜBNER Photonics, Sweden) for optogenetic stimulation. The power of the laser was programmed by the software and fine-tuned by a continuous filter (NDC-50C-2M, Thorlabs, NJ, USA) to 10 mW at the end of the patch cord (~ 71.59 mW/mm² at the end of the implanted fiber). The stimulation timing was controlled by a shutter (SRS470, Stanford Research System, Sunnyvale, CA) and a Master-8 stimulus generator (A.M.P.I., Jerusalem, Israel). In avoidance and approach, a total of 72 trials were divided into 12 alternating blocks (OFF-ON-OFF...), and 2-second continuous stimulation was delivered 0.5 seconds after the onset of a tone during stimulation blocks. In the open field test, a 32-minute test was divided into eight alternating stimulation blocks (OFF-ON-OFF...), and a 2-second continuous stimulation was delivered every 40 seconds during the stimulation blocks.

For optogenetics animals, eight experimental and five controls were tested in the following order: open field test under dim light, reward approach, active avoidance, open field test under bright light.

Reward Approach (Optogenetics)

Animals were trained using the same conditioning and chamber and tested in the same chamber as mentioned in *Reward Approach (Photometry)*. After animals reached the learning criteria (70% success rate with 5-second window), animals were then trained with a patch cord attached for another one to two days to habituate the animals to a patch cord. On the test day, 72 trials were divided into 12 alternating blocks (OFF-ON-OFF.....), and a 2-second continuous stimulation was delivered 0.5 seconds after the onset of tone during stimulation blocks, regardless of the behavioral outcome.

Active avoidance (Optogenetics)

Animals were trained using the same conditioning and chamber, and were tested in the same chamber as mentioned in *Behavior Paradigm: Active Avoidance (Photometry)*. After animals reached the learning criteria (70% success rate with 5-second window), animals were then trained with a patch cord attached for another 1-2 days to habituate the animals to a patch cord. On the test day, 72 trials were divided into 12 alternating blocks (OFF-ON-OFF.....), and a 2-second continuous stimulation was delivered 0.5 seconds after the onset of tone during stimulation blocks, regardless of the behavioral outcome.

Open Field Test (Optogenetics, Dim light)

A 26 W x 48 L x 21 H (cm) clean rectangular rat homecage with mouse homecage bedding placed in a sound-proof box (MedAssociates, Fairfax, VT) lit by a red LED strip was used for an open field test. Mice were first habituated with their cagemates, food, and water in the arena for an hour the day before testing. During the test, food and water were removed from the arena, and each mouse was tested individually. At the start of the experiment, mice were first connected to the patch cord fiber and then placed in the center of the arena. A 32-minute test was divided into eight alternating stimulation blocks (OFF-ON-OFF.....), a 2-second continuous stimulation was delivered every 40 seconds during the stimulation blocks.

Open Field Test (Optogenetics, Bright light)

A 46 W x 46 L x 30.5 H (cm) white rectangular box made with PVC was used for an open field test under bright room light. Mice were first connected to the patch cord fiber and then placed in the center of the arena at the start of the experiment. A 32-minute test was divided into eight alternating stimulation blocks (OFF-ON-OFF.....), a 2-second continuous stimulation was delivered every 40 seconds during the stimulation blocks.

Perfusion and Histology Verification

After experiments, animals were deeply anesthetized with pentobarbital at a dose of 90 mg/kg and perfused with 20 ml PBS, followed by 20 ml 4% paraformaldehyde solution. Brains were soaked in 4 °C 4% paraformaldehyde for 20 hours and then switched to 30% sucrose solution for 20-40 hours until the brains sank. Brains were sectioned coronally (40-50 μ m) with a freezing microtome and then washed with PBS and mounted with PVA-DABCO. Images were acquired using a Zeiss confocal with 5x air, 20x water, and 40x water objectives.

Statistics and Data analysis

All data analysis and statistical testing were performed using custom-written scripts in MATLAB 2019 (MathWorks, Natick, MA). For all behaviors, location and movement were tracked using Ethovision XT10 (Noldus Leesburg, VA).

Error bars and shaded areas in figures report standard error of the mean (s.e.m.). All statistical tests were two-tailed. Within-subject analyses were performed using paired t-test, and between-subject analyses were performed using an unpaired t-test.

Photometry signal analysis

$\Delta F/F$ was calculated with equation (Eq. 1 [↗](#)). The signal measured from the 405 nm reference channel was linearly fitted to the 475 nm signal and was subtracted from the 475 nm signal. The difference was then divided by an exponential function ($a \cdot e^{b\%} + c$) fitted to 475 nm signal.

$$\frac{\Delta F}{F} = \frac{475\text{nm signal} - \text{fitted } 405\text{nm signal}}{\text{exponential decay fitted to } 475\text{nm signal}} \times 100 \text{ (Eq. 1)}$$

Open field test analysis

Mouse speed was first grouped into three clusters, nonmovement, low movement, and high movement, using k-means. The movement threshold was thus defined by the lowest speed of the low movement cluster. Then the peak speed of the movement was detected by finding local maxima with absolute peak value larger than a threshold and at least 5 cm/s larger than the baseline. (This was performed using *findpeak* function in MATLAB and with 'MinPeakProminence' set to 5 and with 'MinPeakHeight' set to movement threshold.) Movement initiation was defined by when the speed first went above 10% of the peak speed within 2 seconds before the epoch. Only epochs with at least 1 second of less than threshold speed before movement initiation were included. Group photometry analyses compared mean $\Delta F/F$ 4 to 2 seconds before and 2 seconds around the peak speed of the movement. Optogenetic analyses compared speed differences of NpHR- and eYFP-expressing mice between mean speed during the 2-second stimulation period and mean speed at the 2-second period right before stimulation.

Approach task & Active Avoidance task

Group Photometry Analysis

Group photometry analyses compared mean $\Delta F/F$ 4 to 2 seconds before and 1 second before and after chamber crossing, and only successful trials where animals crossed the chamber within a 5-second window were included. In the active avoidance task, movement initiation was defined by when the speed first went above 10% of the speed at the chamber crossing within 2 seconds before the chamber crossing. In **Figure 1—Figure Supplement F**, [G](#) [↗](#) and [H](#) [↗](#), the maximal $\Delta F/F$ around crossing was calculated by taking the maximum $\Delta F/F$ from 0.5 seconds before to 1 second after avoidance chamber crossing of each trial, and the corresponding maximal speed around crossing was calculated by taking the maximum speed 0.5 seconds before and after avoidance chamber crossing.

Movement Detection

Movement epochs in habituation and inter-trial intervals in avoidance tasks were detected by finding local maxima with absolute peak value larger than 10 cm/s and at least 3 cm/s larger than the baseline in smoothed speed traces. This was performed using *findpeak* function in MATLAB and with 'MinPeakProminence' set to 3 and with 'MinPeakHeight' set to 10. The speed traces were smoothed with Gaussian-weighted moving average with a window of seven frames by MATLAB

function *smoothdata* before the movement epoch detection. (The video was recorded in 15 frames per second). The movement was excluded if it happened within 5 seconds after another movement. Habituation was defined as the time starting from when the animals were placed in the chamber to the start of the first Pavlovian conditioning trials on the first day of training.

Movement in the intertrial intervals was measured from 5 seconds after the end of the trial to 5 seconds before the start of the next trial in a well-learned session. Movement initiation was defined by when the speed first went above 10% of the peak speed within 2 seconds before the epoch.

Group Optogenetic Speed and Crossing Probabilities Analysis

Optogenetic speed analyses compared speed differences of NpHR- and eYFP-expressing mice during the 2-second stimulation period (0.5 seconds to 2.5 seconds from the tone onset) in stimulation trials to the same period in non-stimulation trials. Trials in which animals crossed before 0.5 seconds after tone onset or before the laser stimulation onset were excluded from speed analyses ([Figure 5 D-E](#), [5H-I](#) and [Figure 5—Figure Supplement A](#) and [C](#)). To plot a histogram of crossing probabilities under optogenetic stimulation, we first binned the crossing latency of all trials into 0.25-second bins ranging from 0 to 7 seconds and then normalized by the total number of trials to obtain crossing probability. Optogenetic crossing probability analyses compared differences between NpHR- and eYFP-expressing mice in crossing probability during the 2-second stimulation period (0.5 seconds to 2.5 seconds from the tone onset) in stimulation trials to non-stimulation trials.

Group Optogenetic Block Structure Analysis

In [Figure 5—Figure Supplement \(I-L\)](#), total of 72 active avoidance trials with 12 off-on alternating stimulation blocks, pattern described in [Figure 5A](#), were sectioned into 6 chunks, each chunk containing 6 trials before stimulation and 6 trials with stimulation. Each data point at each time point is the average latency across 6 chunks of all NpHR-expressing mice. In [Figure 5—Figure Supplement \(K-L\)](#), a total of 72 active avoidance trials with 12 off-on alternating stimulation blocks, pattern described in [Figure 5A](#), were then sectioned into 5 chunks, each chunk containing 6 trials with stimulation and 6 trials after stimulation. Due to the stimulation pattern, first “off” block and last “on” blocks were not included. Each data point at each time point is the average latency across 5 chunks of all NpHR-expressing mice. In [Figure 5—Figure Supplement Q](#), a total of 72 active avoidance trials were sectioned into five chunks; each chunk contains three trials before stimulation, six trials with stimulation, and three trials after stimulation. Each point plotted on the graph is the average across five chunks and of all NpHR-expressing animals.

Freezing Detection

Freezing is detected from video based on changes in pixels by a MATLAB function written by David A. Bulkin and Ryan J. Post. To detect the freezing of animals, we used the code first to convert the video into grayscale, crop the window to include only the bottom of the chamber, and then set the pixels belonging to a mouse to 1 s by thresholding pixel values below 27 out of 255. Once the pixels belonging to a mouse were assigned, the code compared the number of pixels changed between frames to get raw movement data. Raw movement data was then filtered with a bandpass filter between 0.01 and 0.9 to ensure the frequency of switching between freezing and non-freezing states matches with the behavior observed. Freezing was then marked when less than 190 mouse pixels were changed in filtered movement data. For [Figure 5—Figure Supplement H](#), duration of freezing was calculated by the total number of frames when the animals were detected freezing during avoidance latency.

Cross-covariance between photometry signals and speed

Photometry signals were first filtered by a lowpass IIR filter with half-power frequency at 7 Hz and then interpolated to match video recording timeframes (15 Hz). Each segment included photometry and speed data from 5 seconds after tone offset to 5 seconds before the next tone onset. Cross-covariance was calculated with an offset of ± 2 seconds for each segment by the *xcov* function in MATLAB. The cross-covariance value with the largest absolute value within the offset range was selected to be the cross-covariance value of the segment (Seo et al., 2019 [DOI](#)).

Linear regression model

The calcium signal was modeled as a linear combination of characteristic neural responses to each sensory or action event following the scheme of Musall et al. (2019) [DOI](#) and Parker et al. (2016) [DOI](#) (Musall et al., 2019 [DOI](#); Parker et al., 2016 [DOI](#)). These events included tone onset, tone offset, avoidance, escape, shock onset, and chamber crossing in the intertrial intervals. The model can be written as the following equation, where $\Delta F/F(t)$ is the recorded calcium dynamics at time t , k_i and j_i are the kernel coefficients of event type I, 1 is the relative time points in a kernel, n_1 is the total number of action events, and n_2 is the total number of sensory events. Each event kernel k_a or j_b was placed at the time point where the event a or b occurred. For action events, including avoidance, escape, and chamber crossing in intertrial intervals, we used kernels (k_i) ranging from 1 second before to 2 seconds after the event to cover the action from initiation to termination. For sensory events, including tone onsets, tone offsets, and shock onsets, kernels (j_i) started right at the moment where the event happened and lasted for 2 seconds afterwards to model the sensory responses.

$$\Delta F/F(t) = \sum_{i=1}^{n_1} \left(\int_{\tau=-1s}^{2s} k_i(\tau) * (t - \tau) d\tau \right) + \sum_{i=1}^{n_2} \left(\int_{\tau=0s}^{2s} j_i(\tau) * (t - \tau) d\tau \right) + \text{error (Eq. 2)}$$

$\Delta F/F(t)$ was modeled as the linear combination of all event kernels. Given the event occurrence time points of all event types, we can use linear regression to decompose characteristic kernels for each event type. Kernel coefficients of the model were solved by minimizing the mean square errors between the model and the actual recorded signals. To prove that kernel k_i is an essential component for the raw calcium dynamics, we compared the explanation power of the full model to the reduced model where the time points of the occurrence of event k_i were randomly assigned. Thus, the kernel coefficients should not reflect the response to the event in the reduced model. The coefficients of determination (R^2) were compared between the reduced model and the full model to estimate the unique contribution of certain events to the explanation power of the model. More details of the methods can be viewed in (Musall et al., 2019 [DOI](#); Parker et al., 2016 [DOI](#)).

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Competing interest statement

The authors declare no competing interest.

Author contributions

Y.-Y.H. and M.R.W. conceived the idea and designed the experiments. Y.-Y.H., Q.Y. and P.B. initiated and performed the experiments. Y.-Y.H., P.B., D.A.B. and M.R.W. worked on different facets of the data analysis. Y.-Y.H., D.A.B. and M.R.W. wrote the paper, and all authors contributed to the final manuscript.

Data accessibility

The authors declare that the data supporting the findings of this study are available within the paper, the methods section, and Extended Data files. Raw data are available from the corresponding author upon reasonable request.

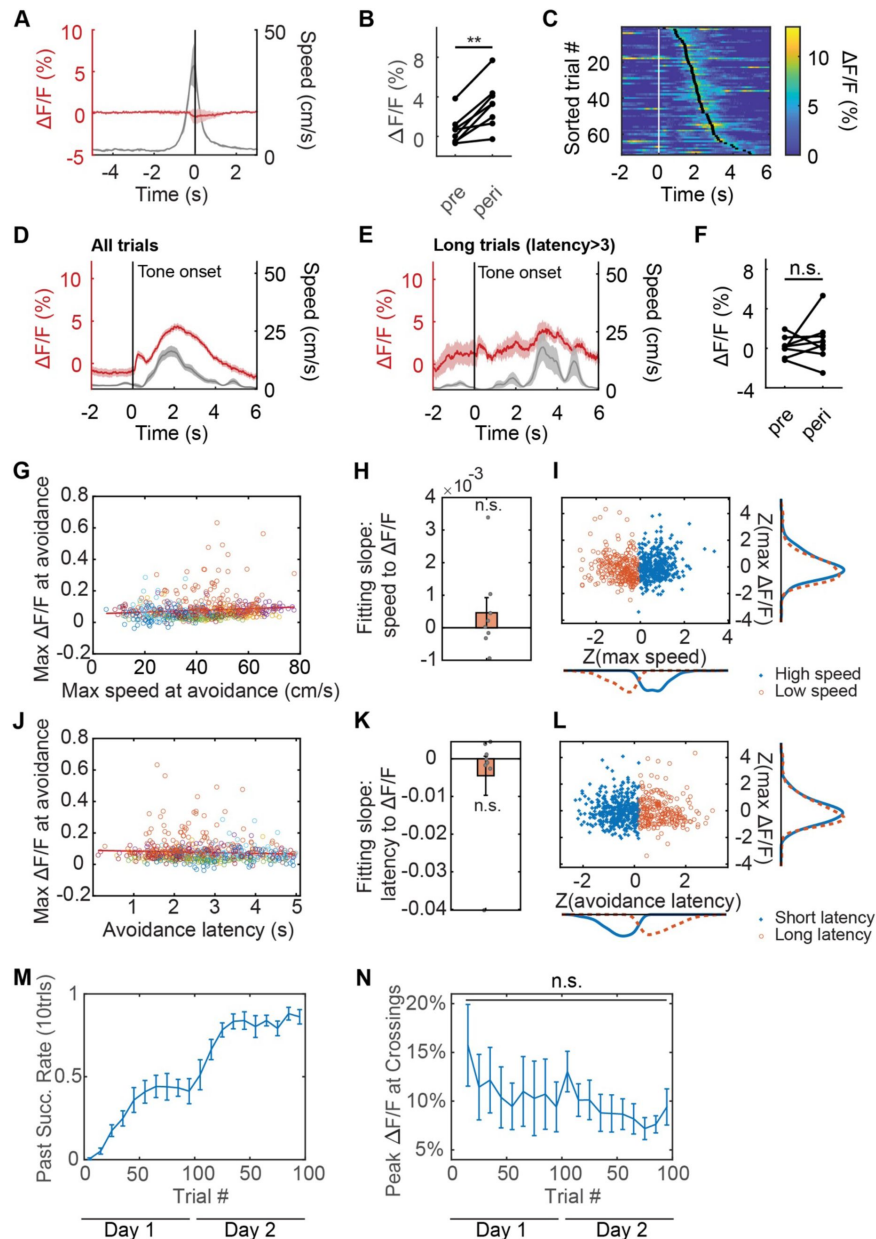


Figure S1—Figure supplement 1.

(A) Average control IL GFP PV $\Delta F/F$ (red) and speed (grey), aligned to chamber crossing in avoidance ($N = 2$ mice). (B) Average IL PV GCaMP6f $\Delta F/F$ before (pre, -2 to 0 seconds) and after (peri, 0 to 2 seconds) avoidance movement initiation. (C) IL PV GCaMP6f $\Delta F/F$ during successful avoidance trials, data aligned to tone onset. White ticks: tone onset. Black ticks: chamber crossing. (D) Example average IL PV GCaMP6f $\Delta F/F$ (red) and speed (grey), aligned to tone onsets. Same example mouse as C. (E) Example average IL PV GCaMP6f $\Delta F/F$ (red) and speed (grey) of long latency trials (avoidance latency longer than 3 seconds), aligned to tone onset. Same example mouse as C. (F) Average IL PV GCaMP6f $\Delta F/F$ before (pre, -4 to -2 seconds) and after (peri, 0 to 2 seconds) tone onset of long latency trials (avoidance latency longer than 3 seconds). (G) Correlation between maximum speed and peak IL PV GCaMP6f $\Delta F/F$ at avoidance. Each dot represents an avoidance trial, and trials from one animal are marked in the same color. (H) Distribution of slope of linear correlation between maximum speed and peak IL PV GCaMP6f $\Delta F/F$ at avoidance. One dot represents one animal. (I) Clustering of distribution of z score of maximum speed and z score of IL PV GCaMP6f $\Delta F/F$ at avoidance. (J-L) The same as (G-I), but the correlation was calculated between avoidance latency and peak IL PV GCaMP6f $\Delta F/F$ at avoidance. (M) Avoidance success rate over time. Success rate was calculated every 10 trials. (N) Peak IL PV GCaMP6f $\Delta F/F$ at avoidance (average $\Delta F/F$ -1 to 1 second around avoidance) every 10 trials over 2 days. ns = non-significant, $**p < 0.01$; for B and F, paired t-test; for H and K, one-sample t-test; for N, one-way ANOVA. Shaded regions and error bars indicate SEM.

Reward approach task

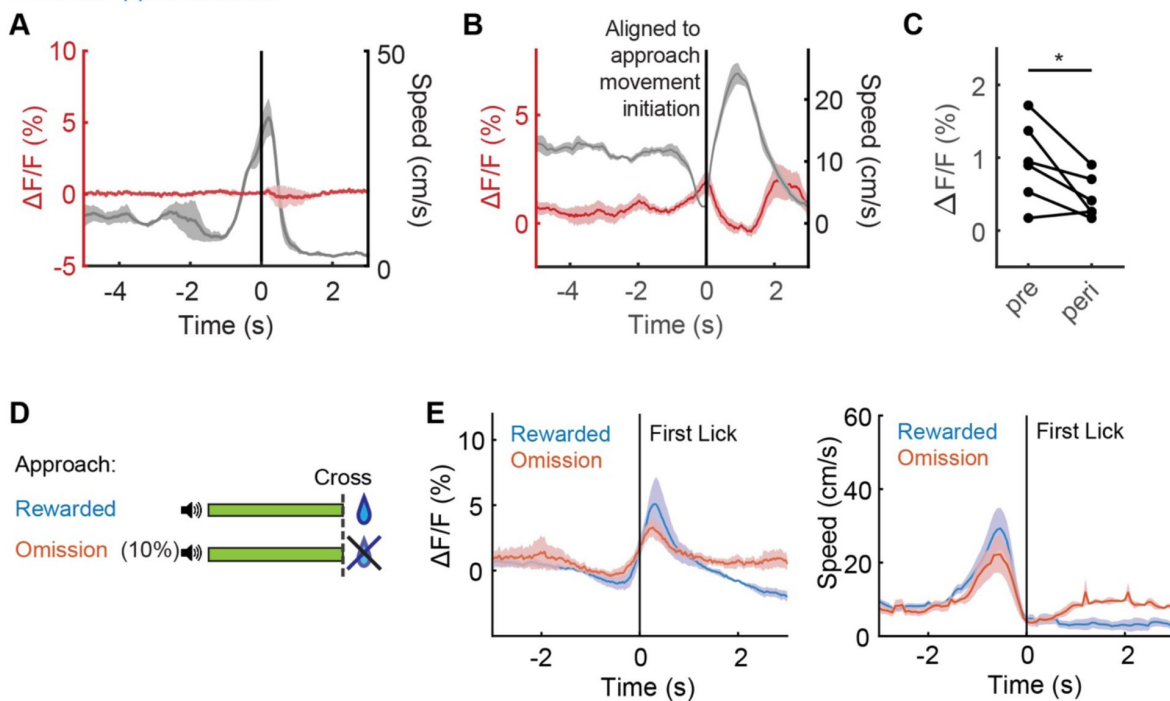


Figure S3—Figure supplement 1.

(A) Average IL PV $\Delta F/F$ (red) and speed (grey), aligned to chamber crossing in approach from PV::GFP mice. (N=2 mice). (B) Average IL PV $\Delta F/F$ (red) and speed (grey), aligned to initiation of movement for reward approach (N = 6 mice). (C) Average IL PV $\Delta F/F$ before initiation of movement for reward approach (pre, 2 to 0 seconds before movement initiation) and after movement initiation (peri, 0 second to 2 seconds after movement initiation) (N = 6 mice, $p=0.0382$, paired t-test). (D) Schematic of a modified reward approach task with reward omissions. (E) Average IL PV $\Delta F/F$ (top) and speed (bottom) of rewarded trials, aligned to the first lick after chamber crossing (N=5 mice).

Open field test

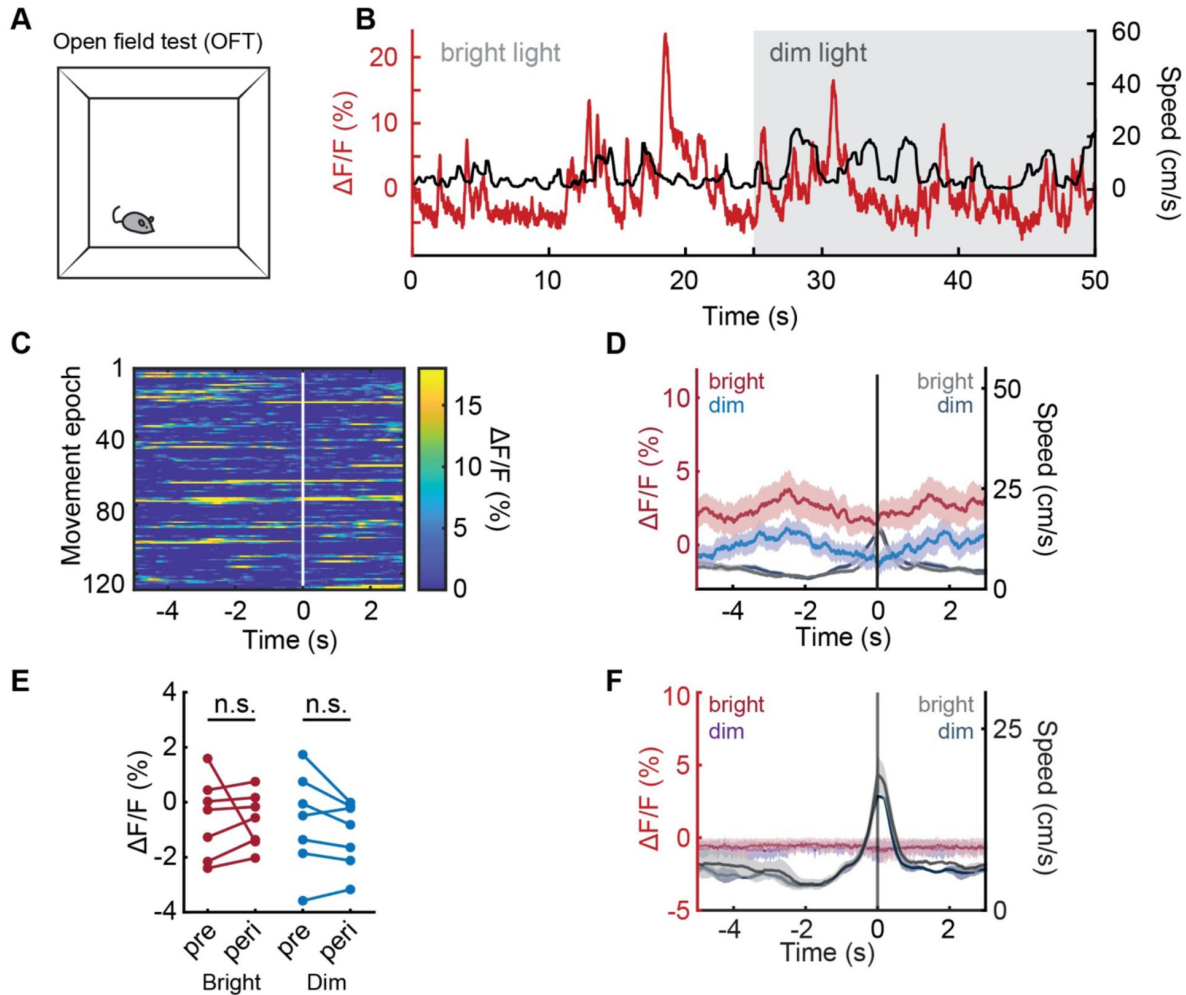


Figure S3—Figure supplement 2.

(A) OFT schematic. (B) Example IL PV GCaMP6f $\Delta F/F$ (red) and speed (black) during OFT under bright light (white shading) and dim light epochs (grey shading). (C) IL PV GCaMP6f $\Delta F/F$ during OFT, aligned to maximum speed of each movement epoch in OFT. White ticks: maximum speed of movement epoch. Same example mouse as B. (D) Example average IL PV GCaMP6f $\Delta F/F$ (red; purple) and speed (grey; dark blue), aligned to maximum speed of each movement epoch in OFT under bright lighting (red; grey) and under dim lighting (purple; dark blue). Same example mouse as B. (E) Average IL PV GCaMP6f $\Delta F/F$ before (pre, -4 to -2 seconds) and during (peri, -1 to 1 seconds) peak movement under bright light (red) and dim light (blue). (F) Average control IL PV GFP $\Delta F/F$ (red; purple) and speed (grey; dark blue), aligned to the maximum speed of each movement epoch in OFT under bright light (red; grey) and dim light (purple; dark blue) (N=2 mice). Shaded regions indicate SEM.

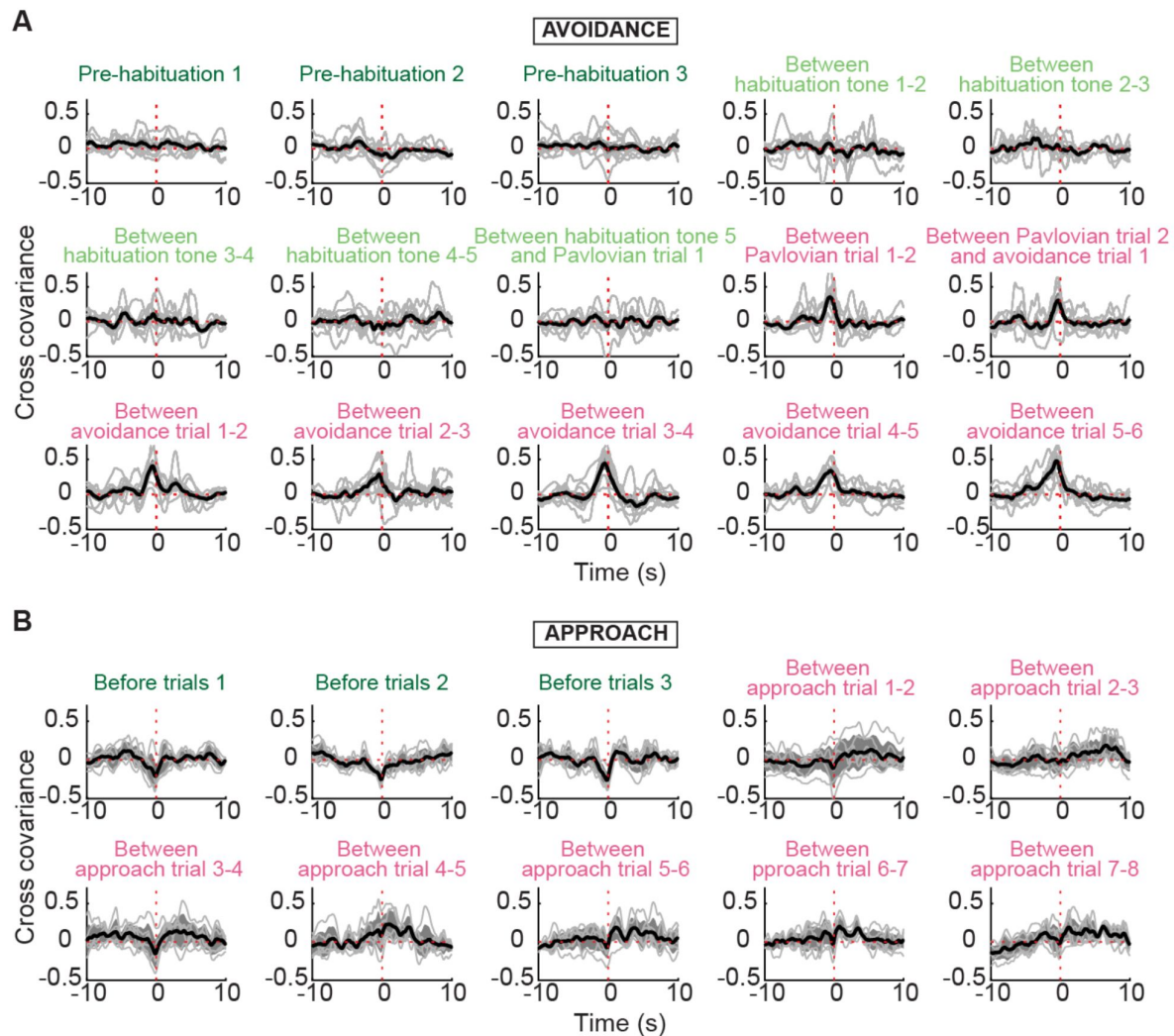


Figure S4—Figure supplement 1.

(A) Evolution of cross-covariance between movement speed and PV activity in habituation and inter-trial intervals on the first day of avoidance training (N=7 mice). (B) Evolution of cross-covariance between movement speed and PV activity in habituation and inter-trial intervals of well-trained animals during the approach task (N=6 mice).

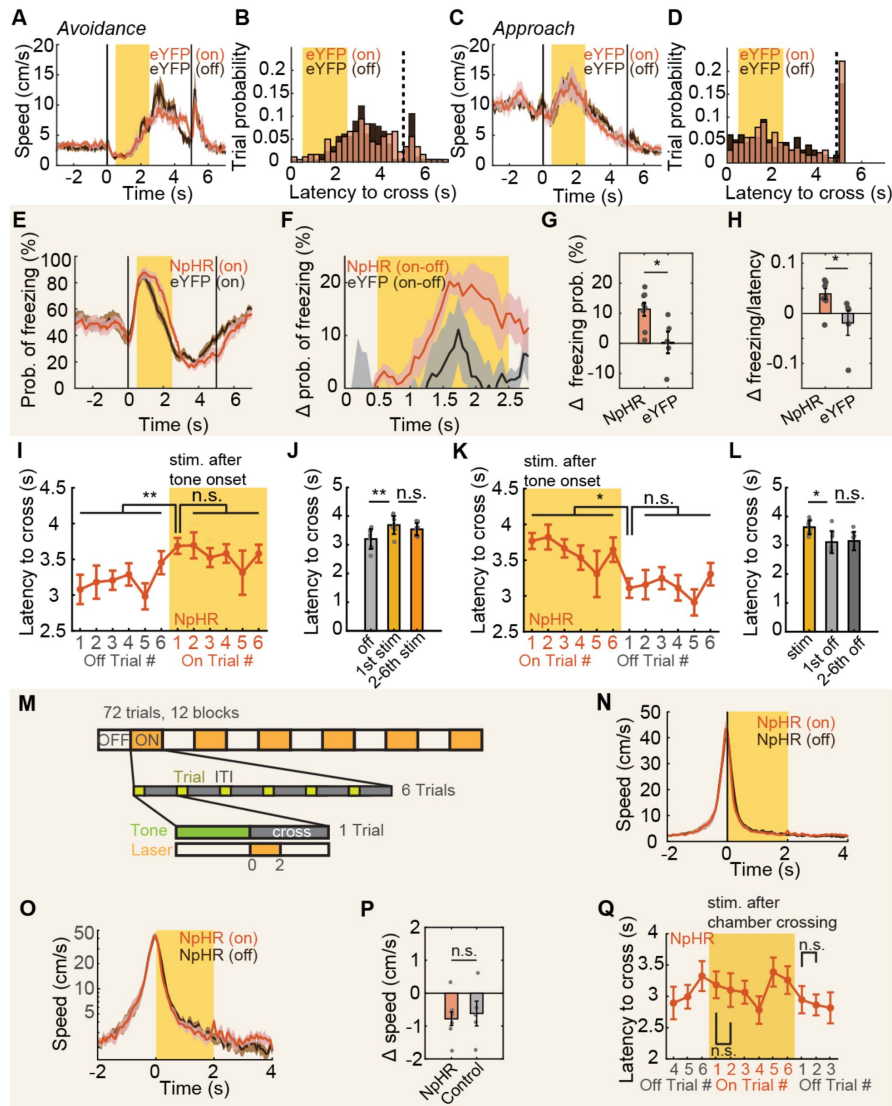


Figure S5—Figure supplement 1.

(A) Speed of eYFP-expressing mice during avoidance with (red) and without (dark brown) illumination. (B) Distribution of crossing latencies during avoidance in eYFP-expressing animals with (red) and without illumination (brown). (C-D) Same as (A-B) for approach. (E) Freezing probabilities of NpHR-expressing animals during avoidance with (red) and without (dark brown) illumination. (F) Difference in freezing probabilities between with (on) and without (off) illumination in NpHR-expressing mice (red) and in eYFP-expression mice (grey). (G) Difference in freezing probabilities, averaged over the laser stimulation period, between with (on) and without (off) illumination in NpHR-expressing mice (red) and in eYFP-expression mice (grey). (H) Change in ratio of freezing duration/avoidance latency of NpHR-expressing and eYFP-expression mice during illumination compared to non-illumination. (I) Crossing latencies during avoidance in NpHR-expressing mice, grouped by relative position prior or within a stimulation block. (J) Average crossing latencies of all non-illuminated trials (off), 1st trials (1st stim) and 2nd-6th trials in illuminated blocks (2-6th stim). (K) Crossing latencies during avoidance in NpHR-expressing mice, grouped by relative position within or after a stimulation block. (L) Average crossing latencies of all illuminated trials (stim), 1st trials (1st off) and 2nd-6th trials (2-6th off) in non-illuminated blocks. (M) Post-crossing optogenetic inactivation schematic. (N) Speed of NpHR-expressing mice during avoidance with (red) and without (dark brown) illumination. (O) Same as (N), but the y-axis was plotted on a log scale. (P) Change in speed of NpHR-expressing and eYFP-expression mice during illumination (0 to 2 s from avoiding chamber crossing) compared to non-illumination trials. (Q) The latency of the first trials in stimulation blocks is compared to the latency of the second trials in stimulation blocks, and the latency of the first trials after stimulation blocks is compared to the latency of the second trials after stimulation blocks. ns = non-significant, * $p < 0.05$, ** $p < 0.01$; for G and H, unpaired t-test; for J, L and Q, paired t-test; for P, 2-way ANOVA interaction term. Shaded regions and error bars indicate SEM.

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Reviewer #2 (Public Review):

Summary:

This study examined the role of a prefrontal cortex cell type in active avoidance behavior. The authors conduct a series of behavioral experiments incorporating fiber photometry and optogenetic silencing. The results indicate that prefrontal parvalbumin (PV) neurons play a permissive role in performing signaled active avoidance learning, for which details are sorely lacking. Notably, infralimbic parvalbumin activity resolves incompatible defensive responses to threat by suppressing conditional freezing in order to permit active instrumental controlling responses. The overall findings provide a significant contribution to our understanding of mechanisms that support aversively motivated instrumental learning and may provide insight into both stress vulnerability and resilience processes.

Strengths:

The writing and presentation of data is clear. The authors use a number of temporally-relevant methods and analyses that identify a novel prefrontal mechanism in resolving the conflict between competing actions (freezing vs escape avoidance). The authors conduct an extensive number of experiments to demonstrate that the uncovered prefrontal mechanism is selective for the initiation of avoidance under threat circumstances, not reward settings or general features of movement.

Weaknesses:

The study exclusively focuses on parvalbumin cells, thus questions remain whether the present findings are specific to parvalbumin or applicable to other prefrontal interneuron subtypes. The exact mechanisms that coordinate infralimbic parvalbumin cell activity and threat avoidance behavior are not explored.

Reviewer #3 (Public Review):

Summary:

Here the authors study the role of parvalbumin (PV) expressing neurons in the ventromedial prefrontal cortex (vmPFC) of mice in active avoidance behavior using fiber photometry and optogenetic inhibition.

Strengths:

The methods are appropriate, the experiments are well done, and the results are all consistent with the conceptual model in which vmPFC PV neurons inhibit freezing to enable avoidance movements. There are good controls to rule out a role for cue offset in triggering changes in PV neuron activity, or for a nonspecific role of vmPFC PV neurons in movement initiation.

Weaknesses:

Although potential mechanisms, i.e., the impact of PV neuron activity on the broader circuit, are discussed, they are not directly examined here. There is some discordance between changes in neural activity and behavior: in Figure 4C, the relationship between PV neuron activity and movement emerges almost immediately during learning, but successful active avoidance emerges much more gradually. Again, this is discussed and plausible explanations for this discrepancy are provided.

<https://doi.org/10.7554/eLife.91221.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Additional Discussion Points

(1) There is not much exploration of potential mechanisms, i.e., the impact of PV neuron activity on the broader circuit. Additionally, the study exclusively focuses on PV cells and does not explore the role of other prefrontal populations, particularly those known to respond to cue-evoked fear states. The discussion should consider how PV activity might impact the broader circuit and whether the present findings are specific to PV cells or applicable to other interneuron subtypes.

We have added an extensive discussion of potential mechanisms and the potential contributions of other interneuron subtypes:

“For example, PV neurons aid in improving visual discrimination through sharpening response selectivity in visual cortex (Lee et al., 2012). In prefrontal cortex, PV neurons are critical for task performance, particularly during performance of tasks that require flexible behavior such as rule shift learning (Cho et al., 2020) and reward extinction (Sparta et al., 2014). Further, PV neurons play an essential role in the generation of cortical gamma rhythms, which contribute to synchronization of selective populations of pyramidal neurons (Sohal et al., 2009; Cardin et al., 2009). Courtin et al (2014) showed that brief suppression of dorsomedial prefrontal (dmPFC) PV neural activity enhanced fear expression, one of the main functions of the dmPFC, by synchronizing the spiking activity of dmPFC pyramidal neurons (Courtin et al., 2014). This result is potentially relevant to our findings, but likely

involves different circuit mechanisms because of the difference in timescale, targeted area, and downstream projection targets (Vertes, 2004). These and other studies support the idea that PV neural activity supports the execution of a behavior by shaping rather than suppressing cortical activity, potentially by selecting among conflicting behaviors by the synchronization of different pyramidal populations (Warden et al., 2012; Lee et al., 2014).

The roles of other inhibitory neural subtypes (such as somatostatin (SOM)-expressing and vasoactive intestinal peptide (VIP)-expressing IL GABA neurons) in avoidance behavior are currently unknown, but are likely important given the role of SOM neurons in gamma-band synchronization (Veit et al., 2017), and the role of VIP neurons in regulating PV and SOM neural activity (Cardin, 2018)."

(2) There is some discordance between changes in neural activity and behavior. For example, in Figure 4C, the relationship between PV neuron activity and movement emerges almost immediately during learning, but successful active avoidance emerges much more gradually. Why is this?

We have added extensive text to the discussion that addresses this issue:

"Interestingly, the rise in IL PV neural activity during movement does not require avoidance learning. IL PV neurons begin to respond during movement immediately after the animal has received a single shock in an environment, but learning to cross the chamber to avoid the signaled shock takes tens of trials. Why is there a discordance between the emergence of the IL PV signal during movement and avoidance learning?

The components underlying active avoidance have been debated over the years, but are thought to involve at least two essential behaviors – suppressing freezing, and moving to safety (LeDoux et al., 2017). Freezing is the default response of mice upon hearing a shock-predicting tone, and can be learned in a single trial (Ledoux, 1996; Fanselow, 2010; Zambetti et al., 2022). When a predator is in the distance, freezing can increase the chance of survival by reducing the chances of detection. However, a strategic avoidance behavior may prevent a future encounter with the predator altogether. The importance of IL PV neural activity in defensive behavior may be to suppress reactive defensive behaviors such as freezing in order to permit a flexible goaldirected response to threat.

The freezing suppression and avoidance movement components of the avoidance response are dissociable, both because freezing precedes avoidance learning, and because animals intermittently move prior to avoidance learning. Our finding that the rise in PV activity during movement emerges immediately after receiving a single shock, tens of trials before animals have learned the avoidance behavior, suggests that the IL PV signal is associated with the suppression of freezing. Further, IL PV neurons do not respond during movement toward cued rewards because in reward-based tasks there is no freezing response in conflict with reward approach behavior."

(3) vmPFC was defined here as including the infralimbic (IL) and dorsal peduncular (DP) regions. While the role of IL has been frequently characterized for motivated behavior, relatively few studies have examined DP. Perhaps the authors are just being cautious, given the challenges involved in the viral targeting of the IL region without leakage to nearby regions such as DP. But since the optical fibers were positioned above the IL region, it is possible that DP did not contribute much to either the fiber photometry signals or the effects of the optogenetic manipulations. Perhaps DP should be completely omitted, which is more consistent with the definitions of vmPFC in the field.

Yes, we included DP to be cautious as our viral expression sometimes leaks into DP, though the optic fiber targets IL. We have replaced vmPFC with IL throughout the manuscript.

(4) In the Discussion, the authors should consider why PV cells exhibit increased activity during both movement initiation and successful chamber crossing during avoidance. While the functional contribution of the PV signal during movement initiation was tested with optogenetic inhibition, some discussion on the possible role of the additional PV signal during chamber crossing is of interest readers who are intrigued by the signaling of two events. Is the chamber crossing signal related to successful avoidance or learned safety (e.g., see Sangha, Diehl, Bergstrom, Drew 2020)?

IL PV neural activity starts to increase at movement initiation, peaks at chamber crossing (when movement speed is highest), and decreases after chamber crossing (Figure 1E). Thus, the increase in PV neural activity at movement initiation and at chamber crossing are different phases of the same event.

We think this signal is unlikely to be a safety signal, and have added text to the discussion to clarify this issue:

“We think the IL PV signal is unlikely to be a safety signal (Sangha et al., 2020). First, the PV signal rises during movement not only in the avoidance context, but during any movement in a “threatening” context (i.e. a context where the animal has been shocked). For example, PV neural activity rises during movement during the intertrial interval in the avoidance task. Further, the emergence of the PV signal during movement happens quickly – after the first shock – and significantly before the animal has learned to move to the safe zone. This suggests a close association with enabling movement in a threatening environment, when animals must suppress a freezing response in order to move. Additionally, the rise in PV activity was specifically associated with movement and not with tone offset, the indicator of safety in this task. Finally, if IL PV neural activity reflects safety signals one would expect the response to be enhanced by learning, but the amplitude of the IL PV response was unaffected by learning after the first shock.”

(5) The primary conclusion here that PV cells control the fear response should be considered within the context of prior findings by the Herry laboratory. Courtin et al (2014) demonstrated a select role of prefrontal PV cells in the regulation of fear states, accomplished through their control over prefrontal output to the basolateral amygdala. The observations in this paper, which used both ChR2 and Arch-T to address the impact of vmPFC PV activity on reactive behavior, are highly relevant to issues raised both in the Introduction and Discussion.

Courtin et al (2014)’s finding is very important. We did not discuss this paper originally because Courtin et al. is about dmPFC, which has a different role in fear processing than IL/vmPFC. We have added text about this finding to the discussion:

“Courtin et al (2014) showed that brief suppression of dorsomedial prefrontal (dmPFC) PV neural activity enhanced fear expression, one of the main functions of the dmPFC, by synchronizing the spiking activity of dmPFC pyramidal neurons (Courtin et al., 2014). This result is potentially relevant to our findings, but likely involves different circuit mechanisms because of the difference in timescale, targeted area, and downstream projection targets (Vertes, 2004).

Additional analyses

(1) As avoidance trials progress (particularly on days 2 and 3), do PFC PV responses attenuate? That is, does continued unreinforced tone presentations lead to reduced reliance of PV cellmediated suppression in order for successful avoidance to occur?

We added Figure 1—Figure supplement 1M and 1N and a sentence on page 5: “IL PV neural activity during the avoidance movement was not attenuated by learning or repeated reinforcement (Figure 1—Figure supplement 1M and N, N = 8 mice, $p = 0.8886$, 1-way ANOVA).” We only included data from days 1 and 2, since we started to introduce short and long tone trials on day 3 which might interfere.

(2) In Figure 3D, it would be very informative and further support the claim of "no role for movement during reward" if the response of these cells during the "initiation of movement during reward-approach" was shown (similar to Figure 1F for threat avoidance).

Thank you for the question. We added Figure 3—Figure supplement 1B and C to show IL PV neural activity aligned to initiation of movement during reward-approach. IL PV activity decreased after movement initiation for reward approach (N = 6 mice, $p = 0.0382$, paired t-test). This further solidifies our claim that IL PV neuron activity only increases for threat avoidance.

Reviewer 1 (Recommendations For The Authors):

(1) Fig1G shows the average response of PV cells during chamber crossing on an animal-to-animal basis. It would be informative to also see a similar plot for movement initiation.

We have added the suggested figure in Figure 1—Figure supplement 1B.

(2) In the Results section (Page 5), there is a small issue with the logic. It says: "As vmPFC inactivation impairs avoidance behavior, the activity of inhibitory vmPFC PV neurons might be predicted to be low during successful avoidance trials." As opposed to "low", it should say "high", right? If inhibition impairs avoidance, then high responding by these cells would be presumed to drive the avoidance response, as supported by your findings.

We have re-worded the text in this section. Based on prior findings that IL inactivation impairs avoidance (Moscarello et al., 2013), we predicted that inhibitory PV neurons would be less active during avoidance, because activating these neurons could suppress IL. However, we found that they were selectively active during avoidance.

(3) In the caption/legend for Fig1E, it says that the "black ticks" indicate "tone onset". But it should say "movement initiation".

We thank the reviewer for pointing out this error. The ticks do indicate tone onset, and we have corrected the figure to reflect this.

Reviewer 2 (Recommendations For The Authors):

(4) Perhaps replace the term 'good outcomes' with 'reinforcing outcomes' or simply 'reinforcement'.

Thank you for the suggestion. We have replaced ‘good outcomes’ with ‘reinforcing outcomes’.

Reviewer 3 (Recommendations For The Authors):

(5) It would be useful to provide some (perhaps speculative) explanation for the discordance between the PV activity-movement relationship and success of active avoidance in Fig. 4C

We have added text to the discussion that addresses this issue:

“Interestingly, the rise in IL PV neural activity during movement does not require avoidance learning. IL PV neurons begin to respond during movement immediately after the animal has received a single shock in an environment, but learning to cross the chamber to avoid the signaled shock takes tens of trials. Why is there a discordance between the emergence of the IL PV signal during movement and avoidance learning?

The components underlying active avoidance have been debated over the years, but are thought to involve at least two essential behaviors – suppressing freezing, and moving to safety (LeDoux et al., 2017). Freezing is the default response of mice upon hearing a shock-predicting tone, and can be learned in a single trial (Ledoux, 1996; Fanselow, 2010; Zambetti et al., 2022). When a predator is in the distance, freezing can increase the chance of survival by reducing the chances of detection. However, a strategic avoidance behavior may prevent a future encounter with the predator altogether. The importance of IL PV neural activity in defensive behavior may be to suppress reactive defensive behaviors such as freezing in order to permit a flexible goaldirected response to threat.

The freezing suppression and avoidance movement components of the avoidance response are dissociable, both because freezing precedes avoidance learning, and because animals intermittently move prior to avoidance learning. Our finding that the rise in PV activity during movement emerges immediately after receiving a single shock, tens of trials before animals have learned the avoidance behavior, suggests that the IL PV signal is associated with the suppression of freezing. Further, IL PV neurons do not respond during movement toward cued rewards because in reward-based tasks there is no freezing response in conflict with reward approach behavior.”

(6) I don't really understand what is shown in Figure 4D -- exactly what time points does this represent? Was habituation performed everyday?

Figure 4D shows data from the approach task, not the avoidance task. This data is from welltrained mice, not the first day of training on this task. There was a pre-task recording period every day.

(7) Why was optogenetic inhibition only delivered from 0.5-2.5 sec after the tone cue?

We wanted to avoid any possibility that perception of the tone would be disrupted, so we delayed the onset of optogenetic inhibition. We chose 0.5 sec onset because animals typically begin to move ~1 second after tone onset.

(8) The regression analysis with shuffled time points is not well explained -- some additional methodological details are needed (Fig. 2H).

We added the following to the methods section to provide a clearer explanation:

“DF/F (t) was modeled as the linear combination of all event kernels. Given the event occurrence time points of all event types, we can use linear regression to decompose characteristic kernels for each event type. Kernel coefficients of the model were solved by minimizing the mean square errors between the model and the actual recorded signals. To prove that kernel k_i is an essential component for the raw calcium dynamics, we compared the explanation power of the full model to the reduced model where the time points of the occurrence of event k_i were randomly assigned. Thus, the kernel coefficients should not reflect the response to the event in the reduced model.

Editor's notes:

- Should you choose to revise your manuscript, please include full statistical reporting including exact p-values wherever possible alongside the summary statistics (test statistic and df) and 95% confidence intervals. These should be reported for all key questions and not only when the pvalue is less than 0.05.

Thank you for pointing this out. We have included all the test statistics and exact p values as suggested.

- Please note the sex of the mice and distribution of sexes in each group for each experiment.

We have added the sex of mice for all experiments in the methods section.

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